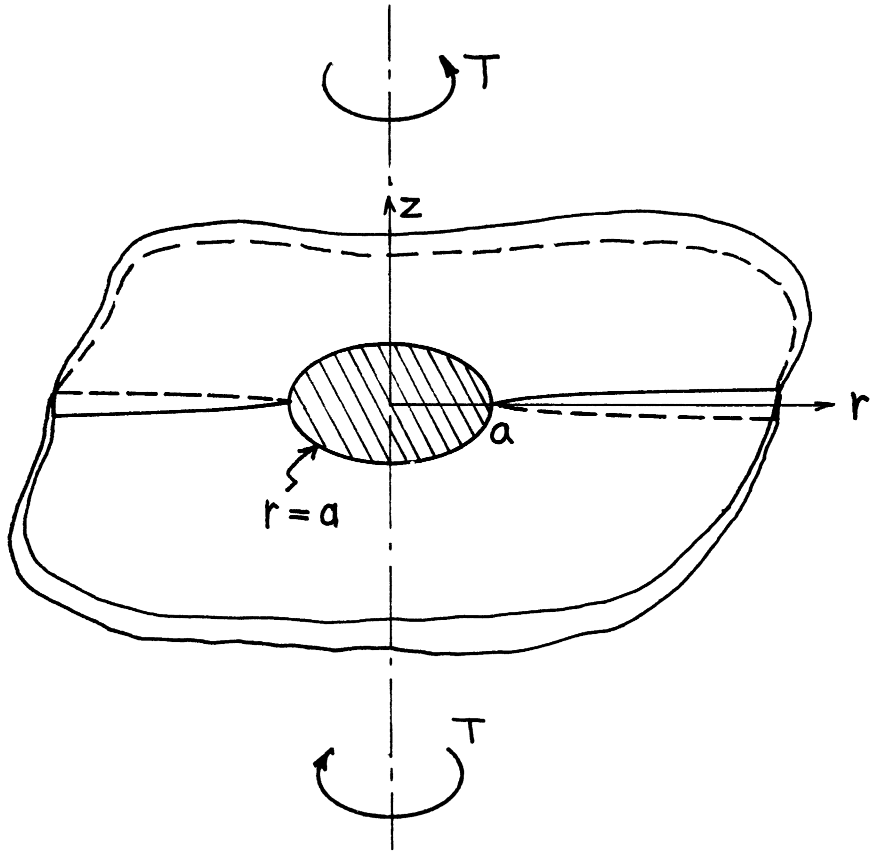
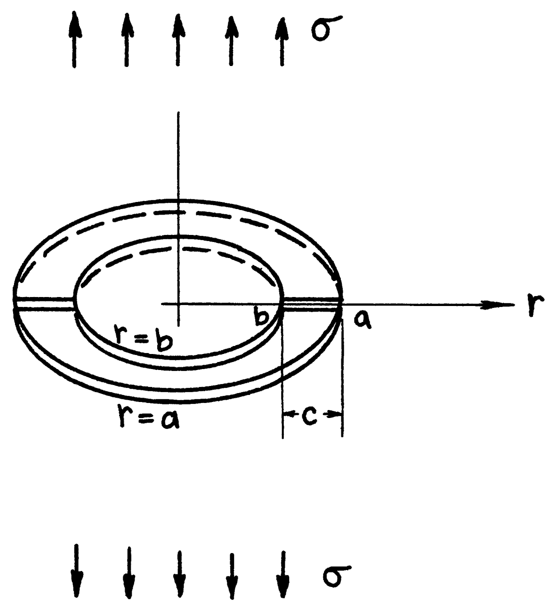


$$K_{III} = \frac{3}{4}\frac{T}{a^2\sqrt{\pi a}}$$
$$= \frac{3}{8}\tau_0\sqrt{\pi a} \qquad \left(\tau_0 = \frac{2T}{\pi a^3}\right)$$

$$K_I = K_{II} = 0$$

Methods: Stress Concentration Factor (Neuber), Solution for a Stamp Problem (e.g., Sneddon)
Accuracy: Exact
References: **Neuber 1937; Sneddon 1951**



$$\underset{r=a}{K_I} = \sigma\sqrt{\pi\frac{c}{2}}\cdot F_a\left(\frac{c}{a}\right)$$

$$\underset{r=b=a-c}{K_I} = \sigma\sqrt{\pi\frac{c}{2}}\cdot F_b\left(\frac{c}{a}\right)$$

where

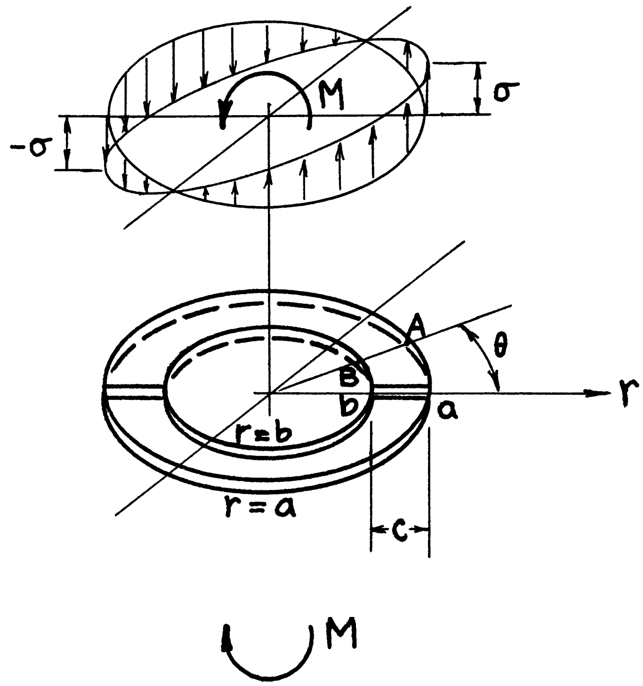
$$F_a\left(\frac{c}{a}\right) = 1 - .116\,\frac{c}{a} + .016\left(\frac{c}{a}\right)^2$$

$$F_b\left(\frac{c}{a}\right) = \frac{1 - .36\frac{c}{a} - .067\left(\frac{c}{a}\right)^2}{\sqrt{1-\frac{c}{a}}}$$

$$\left[F_a\left(\frac{c}{a}\right)\right]_{\frac{c}{a}\to 1} = \frac{2\sqrt{2}}{\pi} = .9002$$

$$\left[\sqrt{1-\frac{c}{a}}\,F_b\left(\frac{c}{a}\right)\right]_{\frac{c}{a}\to 1} = \frac{4\sqrt{2}}{\pi^2} = .573$$

Method: Singular Integral Equation
Accuracy: Better than 0.5% (formulas are based on Erdogan's numerical results)
References: **Erdogan 1982; Tada 1985**



$$\underset{(r=a,0)}{K_{IA}} = \sigma\sqrt{\pi\frac{c}{2}}\cos\theta \cdot F_A\left(\frac{c}{a}\right)$$

$$\underset{(r=b=a-c,\theta)}{K_{IB}} = \sigma\sqrt{\pi\frac{c}{2}}\cos\theta \cdot F_B\left(\frac{c}{a}\right)$$

where

$$F_A\left(\frac{c}{a}\right) = 1 - .401\frac{c}{a} - .065\left(\frac{c}{a}\right)^2 + .066\left(\frac{c}{a}\right)^3$$

$$F_B\left(\frac{c}{a}\right) = \sqrt{1-\frac{c}{a}}\left[.573 + .427\left(1-\frac{c}{a}\right)^{1/4} - .26\left(\frac{c}{a}\right)^{5/2}\left(1-\frac{c}{a}\right)^{3/2}\right]$$

$$\left[F_A\left(\frac{c}{a}\right)\right]_{\frac{c}{a}\to 1} = \frac{4\sqrt{2}}{3\pi} = .6002$$

$$\left[\frac{F_B\left(\frac{c}{a}\right)}{\sqrt{1-\frac{c}{a}}}\right]_{\frac{c}{a}\to 1} = \frac{4\sqrt{2}}{\pi^2} = .573$$

Method: Singular Integral Equation
Accuracy: Better than 0.5% (formulas are based on Erdogan's numerical results)
References: **Erdogan 1982; Tada 1985**

Notes on Solutions for Elliptical Crack Problems

See **pages 26.2, 26.3, and 26.4** (Internal Cracks), and **page 26.5** (External Crack).

1. The angle θ is the parametric angle representing Point A on the crack front. That is, the coordinates of Point A are $[a\cos\theta,\ b\sin\theta,\ (0)]$, as shown in **Fig. 1**.
2. Note that

$$b\left(\sin^2\theta + \frac{b^2}{a^2}\cos^2\theta\right)^{1/2} = \ell$$

is the length of the normal at Point A, as shown in **Fig. 2**.

3. A term $\left(\sin^2\theta + \frac{b^2}{a^2}\cos^2\theta\right)^{1/4}$ repeatedly appears in the solution. By replacing this term with $\sqrt{\ell/b}$, solutions may be expressed more concisely. For example,

$$K_{IA} = \frac{\sigma\sqrt{\pi\ell}}{E(k)}$$

for **page 26.2** and

$$K_{IA} = \frac{P}{2a\sqrt{\pi\ell}}$$

for **page 26.5**.

4. The numerical values of the complete elliptic integrals of the first kind, $K(k)$, and the second kind, $E(k)$, are tabulated in **Appendix L**. Accurate empirical formulas for the second complete elliptic integral, $E(k)$, are also given in **Appendix L**.

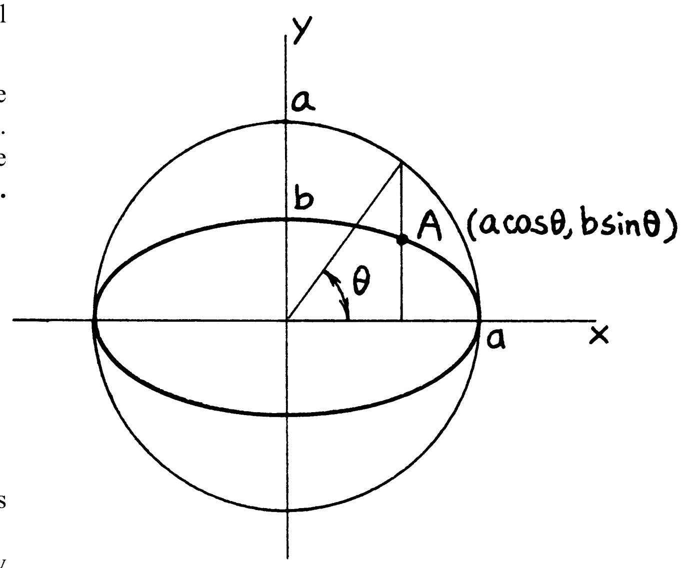


Fig. 1

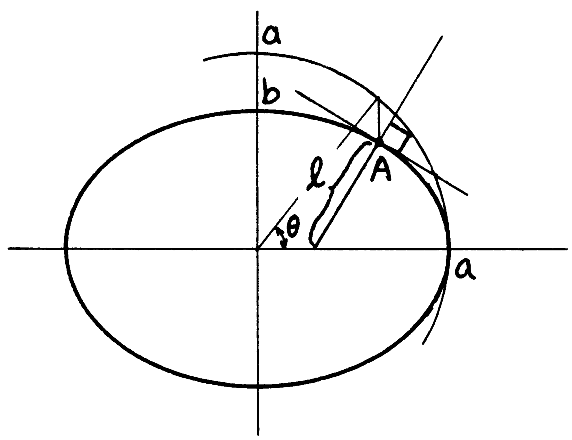
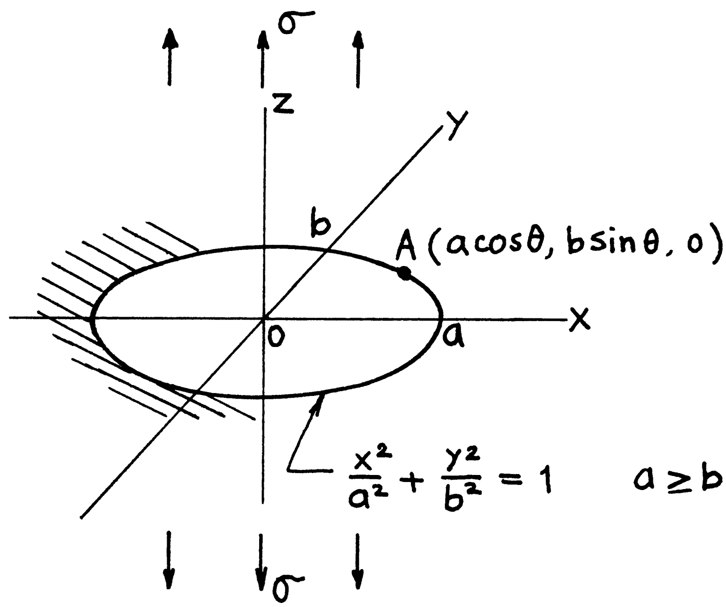


Fig. 2



$$K_{IA} = \frac{\sigma\sqrt{\pi b}}{E(k)}\left\{\sin^2\theta + \frac{b^2}{a^2}\cos^2\theta\right\}^{1/4} = \frac{\sigma\sqrt{\pi \ell}}{E(k)}$$

$$E(k) = \int_0^{\pi/2}\sqrt{1-k^2\sin^2\varphi}\ \ d\varphi$$

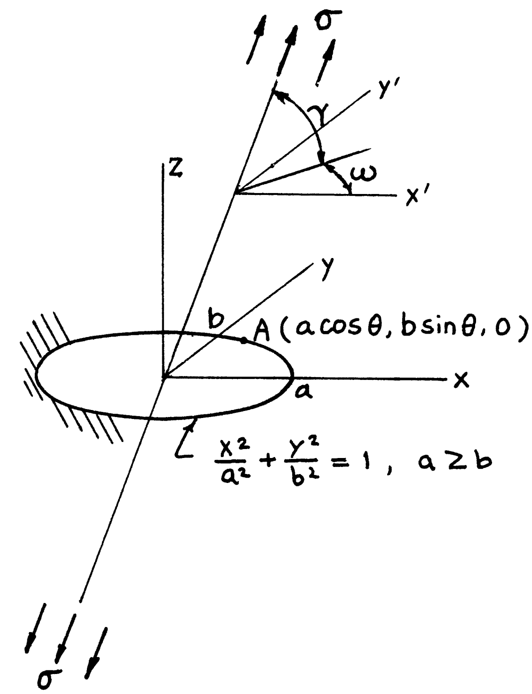
$$k^2 = 1 - b^2/a^2$$

$$K_{I,\max} = K_I\left(\theta = \pm\pi/2\right) = \frac{\sigma\sqrt{\pi b}}{E(k)}$$

$$K_I(a=b) = \frac{2\sigma}{\pi}\sqrt{\pi a}$$

$$K_I(a \to \infty) = \sigma\sqrt{\pi b}$$

Method: Integral Transform (Three-Dimensional Potential Functions)
Accuracy: Exact
References: **Sadowsky 1949; Green 1950; Irwin 1962b**



$$K_{IA} = \frac{\left(\sigma \sin^2 \gamma\right)\sqrt{\pi b}}{E(k)} \left\{ \sin^2 \theta + \left(\frac{b}{a}\right)^2 \cos^2 \theta \right\}^{1/4}$$

$$K_{IIA} = -\frac{(\sigma \sin\gamma \cos\gamma)\sqrt{\pi b}\, k^2}{\left\{ \sin^2 \theta + \left(\frac{b}{a}\right)^2 \cos^2 \theta \right\}^{1/4}} \left\{ \frac{k'}{B} \cos\omega \cos\theta + \frac{1}{C} \sin\omega \sin\theta \right\}$$

$$K_{IIIA} = \frac{(\sigma \sin\gamma \cos\gamma)\sqrt{\pi b}(1-\nu)\, k^2}{\left\{ \sin^2 \theta + \left(\frac{b}{a}\right)^2 \cos^2 \theta \right\}^{1/4}} \left\{ \frac{1}{B} \cos\omega \sin\theta - \frac{k'}{C} \sin\omega \cos\theta \right\}$$

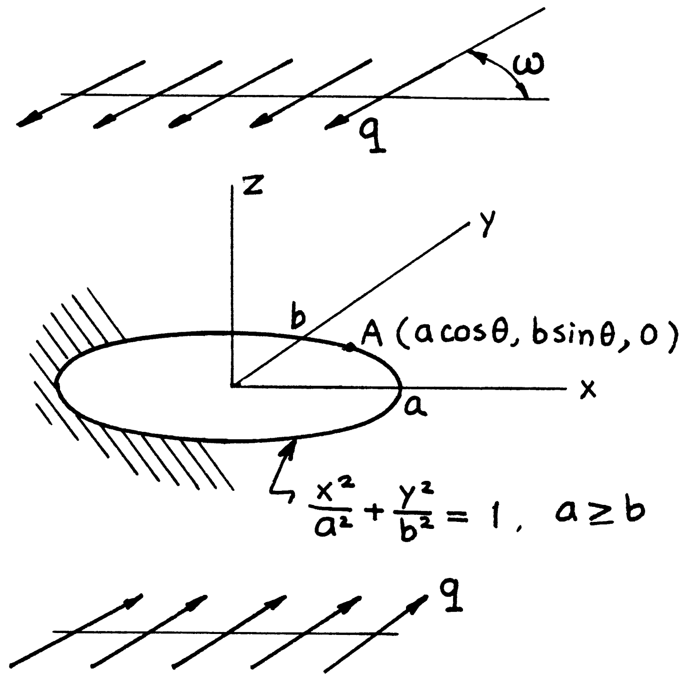
$$B = \left(k^2 - \nu\right) E(k) + \nu k'^2 K(k)$$

$$C = \left(k^2 + \nu k'^2\right) E(k) - \nu k'^2 K(k)$$

$$k^2 = 1 - k'^2, \; k' = b/a$$

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \;\; d\varphi$$

Methods: Three-Dimensional Potential Functions or Superposition of **pages 26.2 and 26.4**
Accuracy: Exact
References: **Kassir 1966; Sih 1968**



$$K_{IIA} = \frac{q\sqrt{\pi b} \cdot k^2}{\left\{\sin^2\theta + \left(\frac{b}{a}\right)^2 \cos^2\theta\right\}^{1/4}} \left\{\frac{k'}{B}\cos\omega\cos\theta + \frac{1}{C}\sin\omega\sin\theta\right\}$$

$$K_{IIIA} = -\frac{q\sqrt{\pi b} \cdot (1-\nu)k^2}{\left\{\sin^2\theta + \left(\frac{b}{a}\right)^2 \cos^2\theta\right\}^{1/4}} \left\{\frac{1}{B}\cos\omega\sin\theta - \frac{k'}{C}\sin\omega\cos\theta\right\}$$

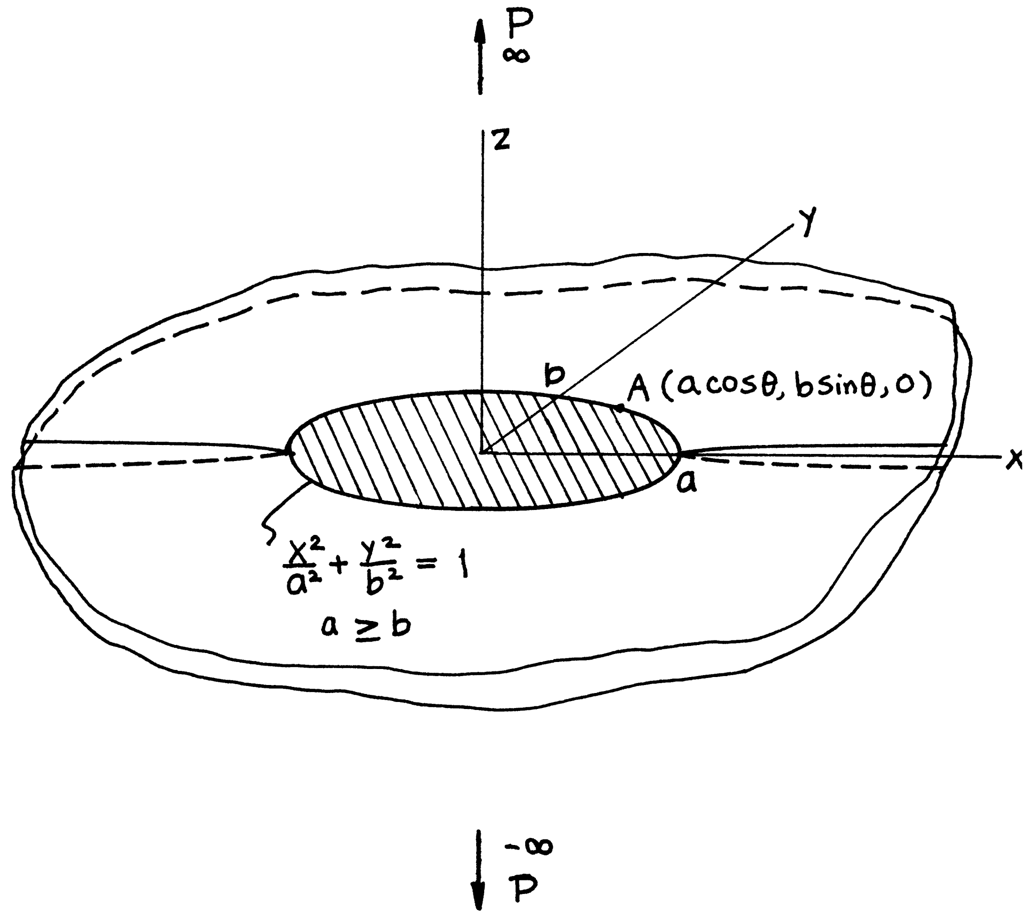
$$B = \left(k^2 - \nu\right)E(k) + \nu k'^2 K(k)$$

$$C = \left(k^2 + \nu k'^2\right)E(k) - \nu k'^2 K(k)$$

$$k^2 = 1 - k'^2, \; k' = b/a$$

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2\varphi}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2\sin^2\varphi} \; d\varphi$$

Method: Three-Dimensional Potential Functions
Accuracy: Exact
References: **Kassir 1966; Sih 1968**



$$K_{IA} = \frac{P}{2a\sqrt{\pi b}} \frac{1}{\left\{\sin^2\theta + \left(\frac{b}{a}\right)^2 \cos^2\theta\right\}^{1/4}} = \frac{P}{2a\sqrt{\pi\ell}}$$

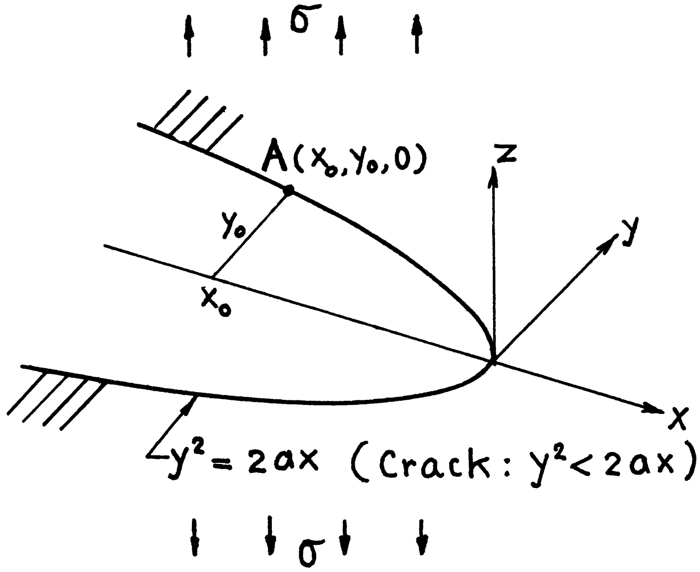
$$K_{IIA} = K_{IIIA} = 0$$

$$K_{I,\ \max} = K_I(\theta = 0) = \frac{P}{2b\sqrt{\pi a}}$$

$$K_I(a = b) = \frac{P}{2a\sqrt{\pi a}}$$

$$K_I(a \to \infty) = \left(\frac{P}{2a}\right)\frac{1}{\sqrt{\pi b}}$$

Method: Fourier Transform (Three-Dimensional Potential Functions)
Accuracy: Exact
References: **Green 1950; Westmann 1966**

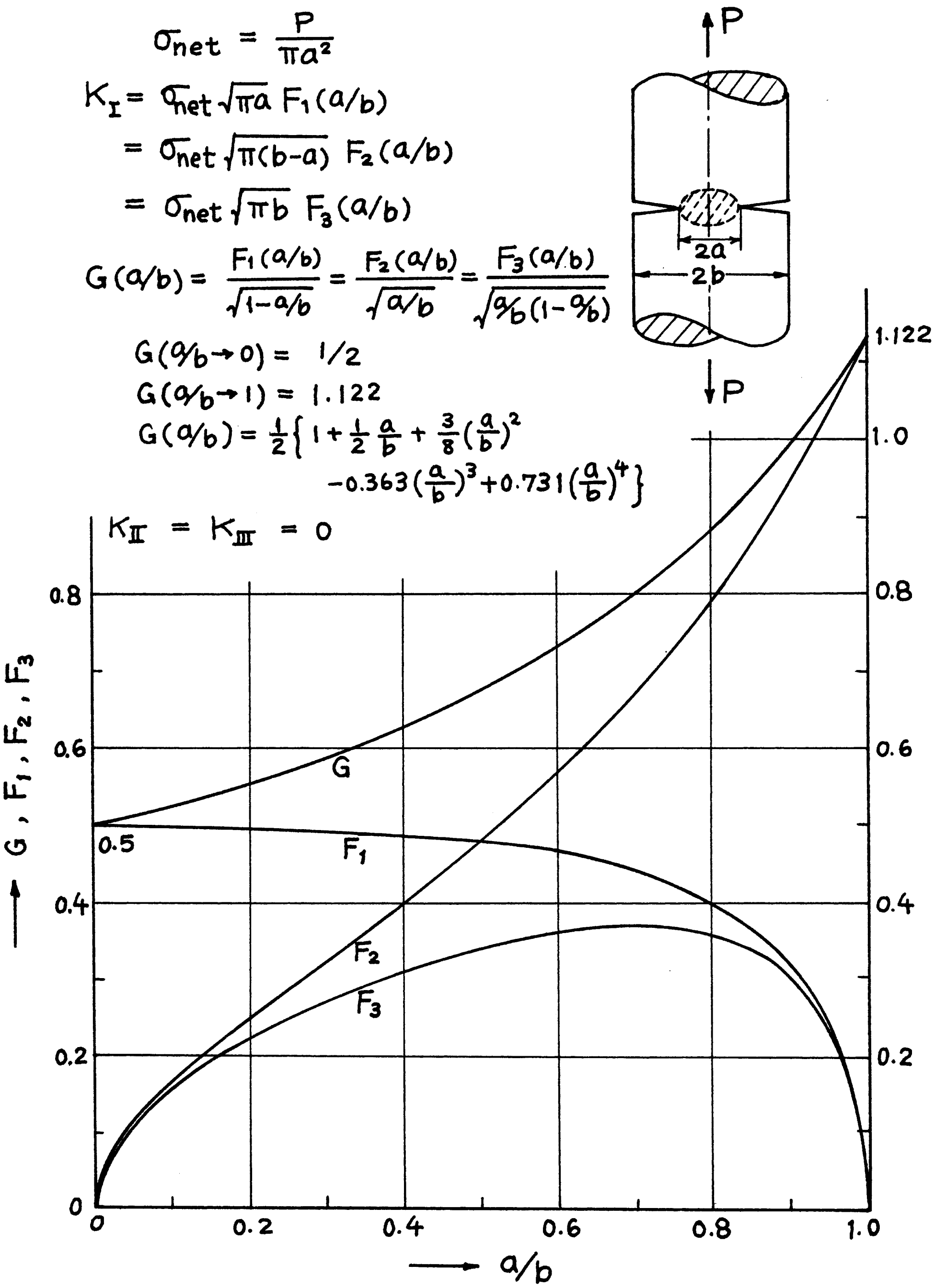


$$K_{IA} = \sigma\sqrt{\pi}\left(a^2 + y_0^2\right)^{1/4}$$

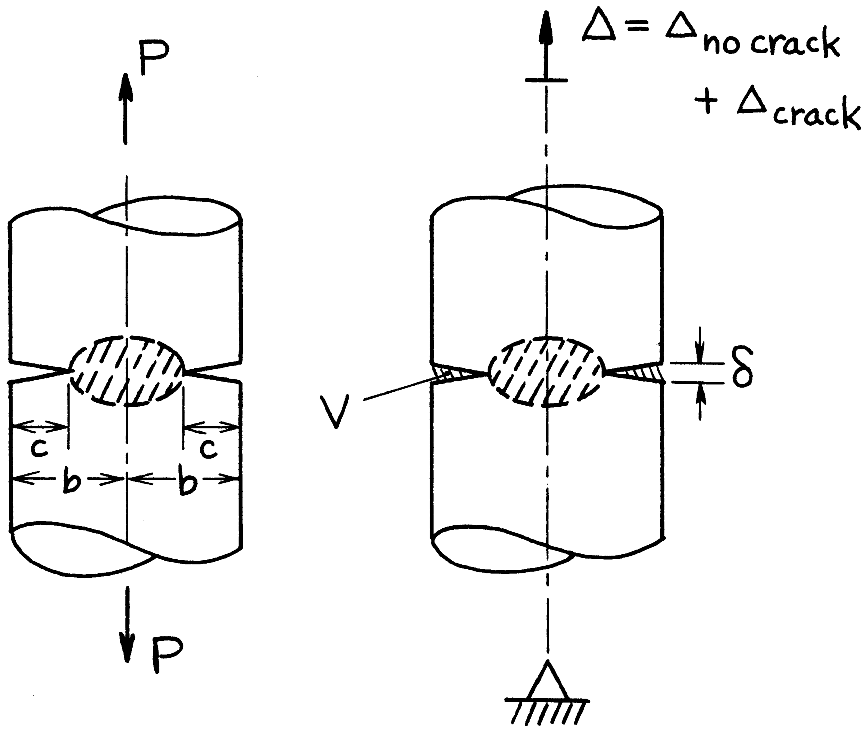
$$= \sigma\sqrt{\pi}\left(a^2 + 2ax_0\right)^{1/4}$$

$$K_{II} = K_{III} = 0$$

Method: Neuber-Papkovich Potential (or a Special Case of **page 26.2**)
Accuracy: Exact
Reference: **Shah 1968**



Method: Singular Integral Equation (Bueckner), Asymptotic Approximation (Benthem)
Accuracy: Better than 1%
Referencees: **Bueckner 1965, 1972; Benthem 1972**
Other References: **Lubahn 1959; Wundt 1959; Irwin 1961; Paris 1965; Zahn 1965; Harris 1967**



$$\sigma = \frac{P}{\pi b^2}$$

$$K_I = \sigma\sqrt{\pi c}\, F(c/b)$$

$$F(c/b) = \frac{1}{(1 - c/b)^{3/2}} \left\{1.122 - 1.302\, \frac{c}{b} + .988\left(\frac{c}{b}\right)^2 - .308\left(\frac{c}{b}\right)^3\right\}$$

Volume of Crack:

$$V = \frac{4(1-\nu^2)}{E}\sigma\pi^2 c^3 G(c/b)$$

Additional Displacement at Infinity due to Crack:

$$\Delta_{crack} = \frac{4(1-\nu^2)}{E}\sigma\pi c\, H(c/b)$$

Crack Opening at Edge:

$$\delta = \frac{4(1-\nu^2)}{E}\sigma c\, D(c/b)$$

where

$$G(c/b) = \frac{1}{3} \cdot \frac{1}{(1 - c/b)^2} \left\{ .375 + .383\left(1 - \frac{c}{b}\right) + .5\left(1 - \frac{c}{b}\right)^3 \right\}$$

$$H(c/b) = (c/b)^2 G(c/b)$$

$$D(c/b) = \frac{1}{(1 - c/b)^2} \left\{ 1.454 - 2.49\frac{c}{b} + 1.155\left(\frac{c}{b}\right)^2 \right\}$$

Methods: K_I, δ Integral Transform ($c/b \leq 0.6$), Interpolation ($c/b > 0.6$); V, Δ Paris' Equation (see **Appendix B**)

Accuracy: K_I, δ 1%; V, Δ 2%

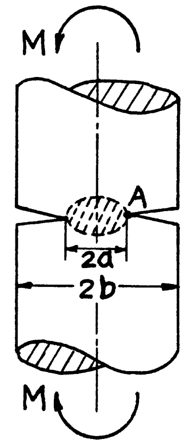
References: **Erdogan 1982; Tada 1985**

NOTE: Δ_{crack} is the elongation at infinity when uniform pressure σ is applied on crack surfaces.

$$\sigma_N = \frac{4M}{\pi a^3}$$

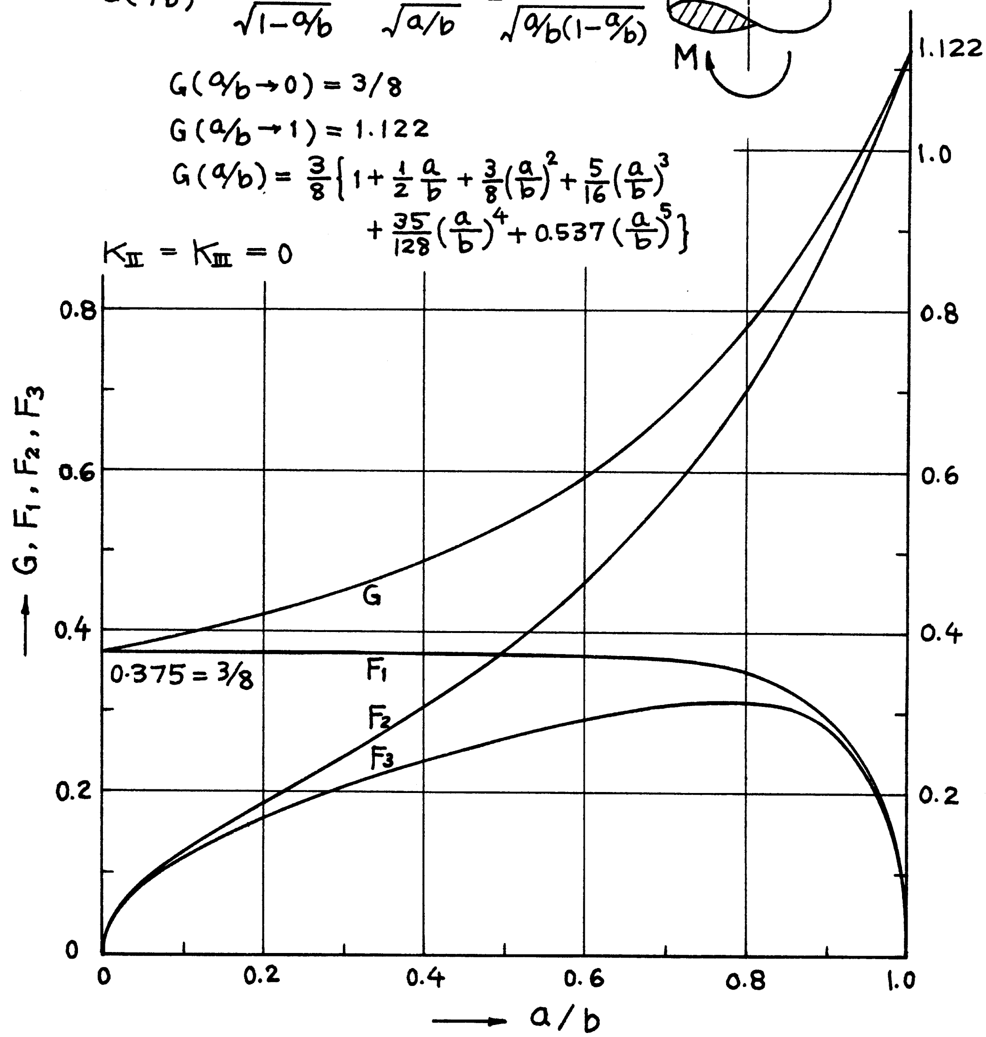
$$K_{I_A} = \sigma_N \sqrt{\pi a}\, F_1(a/b)$$
$$= \sigma_N \sqrt{\pi (b-a)}\, F_2(a/b)$$
$$= \sigma_N \sqrt{\pi b}\, F_3(a/b)$$

$$G(a/b) = \frac{F_1(a/b)}{\sqrt{1-a/b}} = \frac{F_2(a/b)}{\sqrt{a/b}} = \frac{F_3(a/b)}{\sqrt{a/b(1-a/b)}}$$

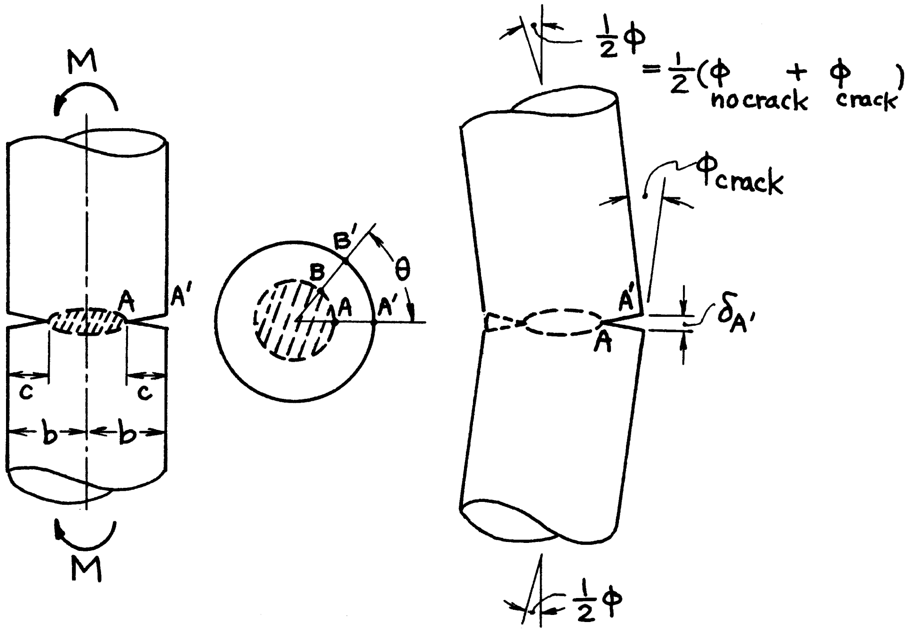


$$G(a/b \to 0) = 3/8$$
$$G(a/b \to 1) = 1.122$$
$$G(a/b) = \frac{3}{8}\left\{1 + \frac{1}{2}\frac{a}{b} + \frac{3}{8}\left(\frac{a}{b}\right)^2 + \frac{5}{16}\left(\frac{a}{b}\right)^3 + \frac{35}{128}\left(\frac{a}{b}\right)^4 + 0.537\left(\frac{a}{b}\right)^5\right\}$$

$$K_{II} = K_{III} = 0$$



Method: Asymptotic Approximation
Accuracy: Better than 1%
Reference: **Benthem 1972**



$$\sigma = \frac{4M}{b^3}$$

$$K_{IA} = \sigma\sqrt{\pi c}\; F(c/b); \quad K_{IB} = K_{IA}\;\cos\theta$$

$$F(c/b) = \frac{1}{\left(1 - c/b\right)^{5/2}}\left\{.563 - .188\frac{c}{b} + \left(1 - \frac{c}{b}\right)^2\left[.559 - 1.47\frac{c}{b} + 2.72\left(\frac{c}{b}\right)^2 - 2.40\left(\frac{c}{b}\right)^3\right]\right\}$$

Additional Rotation at Infinity or Kink at Cracked Section due to Crack:

$$\phi_{crack} = \frac{8\left(1 - \nu^2\right)}{E}\sigma\pi\left(\frac{c}{b}\right)^2\left\{1 - 1.244\frac{c}{b} + 2.11\left(\frac{c}{b}\right)^2 + \frac{.258 - .164\frac{c}{b}}{\left(1 - c/b\right)^3}\right\}$$

Crack Opening at Edge:

$$\delta_{A'} = \frac{4\left(1 - \nu^2\right)}{E}\sigma c\left\{\frac{.454 - .160\frac{c}{b}}{\left(1 - c/b\right)^3} + 1 - 2.6\frac{c}{b}\left(.31 - \frac{c}{b}\right)\left(1 - \frac{c}{b}\right)\left(1 + 3\frac{c}{b}\right)\right\}$$

$$\delta_{B'} = \delta_{A'}\cos\theta$$

Methods: K_I, δ Integral Transform ($c/b \le 0.6$), Interpolation ($c/b > 0.6$); ϕ Paris' Equation (see **Appendix B**)
Accuracy: K_I, δ 1%; ϕ 2%
References: **Erdogan 1982; Tada 1985**

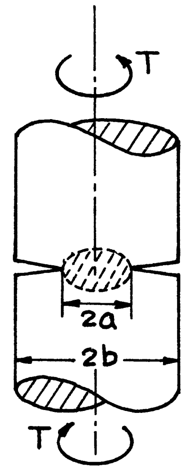
NOTE: Crack surface interference on compressive side is not considered.

$$\tau_N = \frac{2T}{\pi a^3}$$

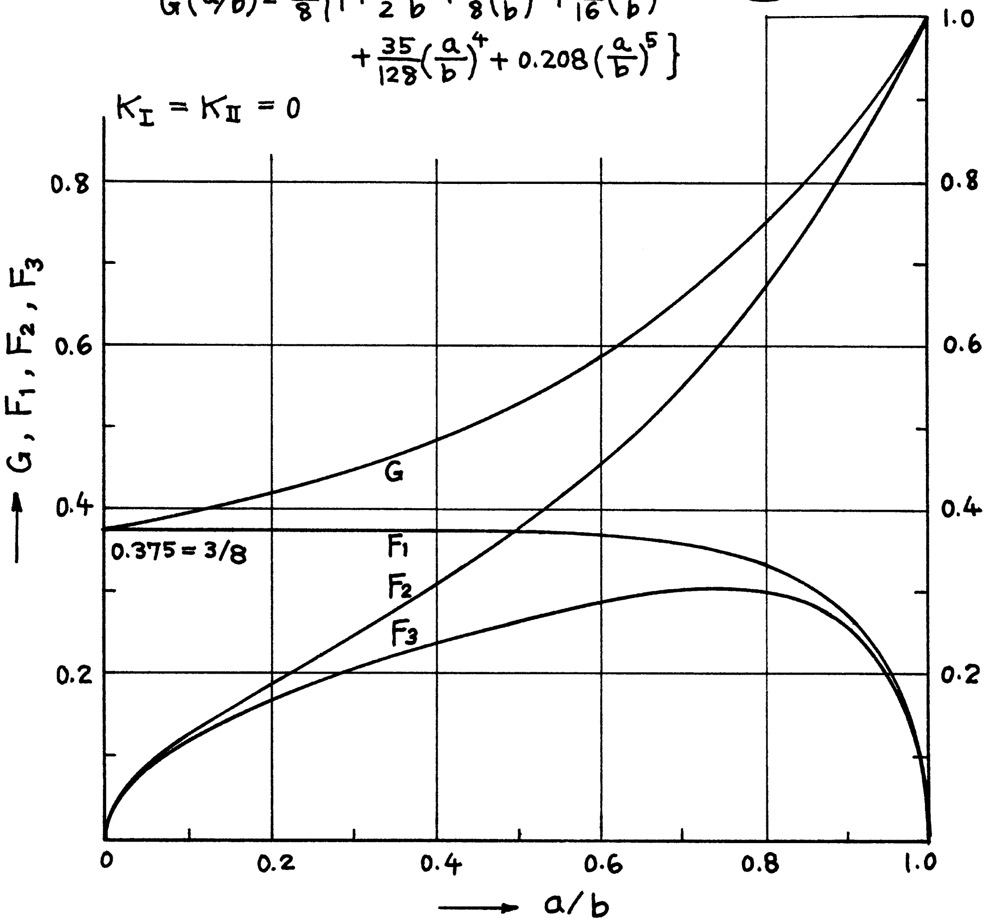
$$K_{III} = \tau_N \sqrt{\pi a}\, F_1(a/b)$$
$$= \tau_N \sqrt{\pi(b-a)}\, F_2(a/b)$$
$$= \tau_N \sqrt{\pi b}\, F_3(a/b)$$

$$G(a/b) = \frac{F_1(a/b)}{\sqrt{1-a/b}} = \frac{F_2(a/b)}{\sqrt{a/b}} = \frac{F_3(a/b)}{\sqrt{a/b(1-a/b)}}$$

$$G(a/b \to 0) = 3/8$$
$$G(a/b \to 1) = 1$$
$$G(a/b) = \frac{3}{8}\left\{1 + \frac{1}{2}\frac{a}{b} + \frac{3}{8}\left(\frac{a}{b}\right)^2 + \frac{5}{16}\left(\frac{a}{b}\right)^3 + \frac{35}{128}\left(\frac{a}{b}\right)^4 + 0.208\left(\frac{a}{b}\right)^5\right\}$$



$K_I = K_{II} = 0$



Method: Asymptotic Approximation
Accuracy: Better than 1%
Reference: **Benthem 1972**

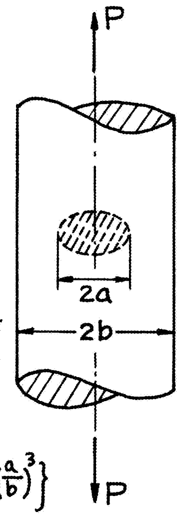
$$\sigma_{net} = \frac{P}{\pi(b^2 - a^2)}$$

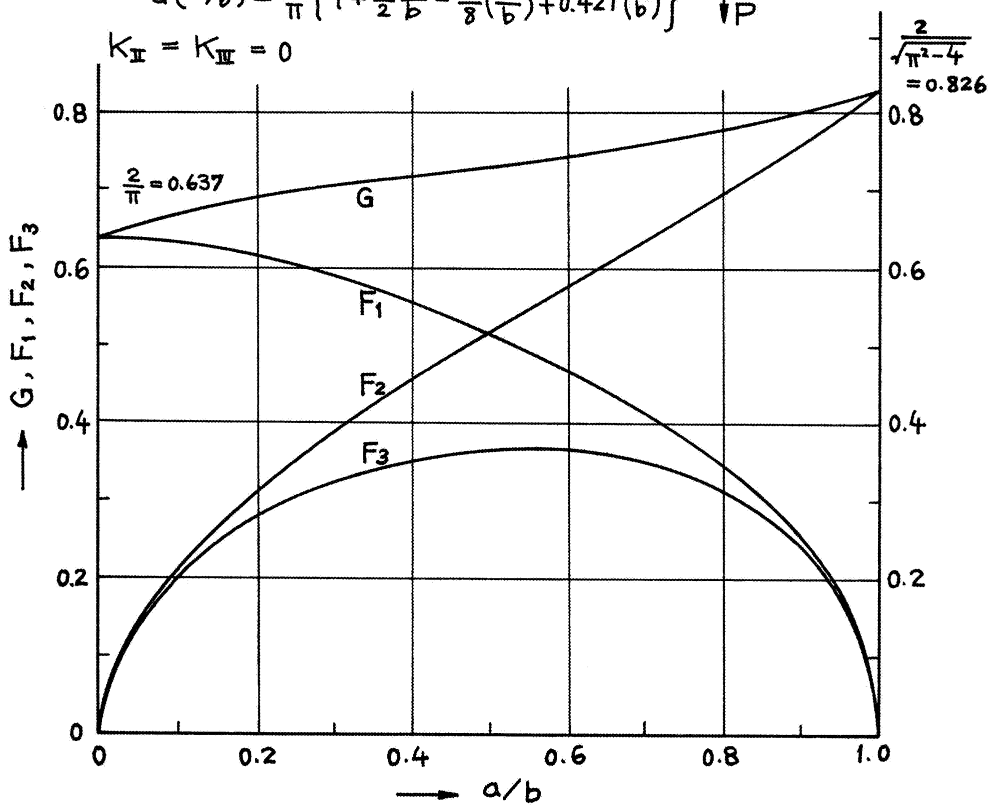
$$K_I = \sigma_{net}\sqrt{\pi a}\,F_1(a/b)$$
$$= \sigma_{net}\sqrt{\pi(b-a)}\,F_2(a/b)$$
$$= \sigma_{net}\sqrt{\pi b}\,F_3(a/b)$$

$$G(a/b) = \frac{F_1(a/b)}{\sqrt{1-a/b}} = \frac{F_2(a/b)}{\sqrt{a/b}} = \frac{F_3(a/b)}{\sqrt{a/b(1-a/b)}}$$

$$G(a/b \to 0) = 2/\pi$$
$$G(a/b \to 1) = 2/\sqrt{\pi^2 - 4}$$
$$G(a/b) = \frac{2}{\pi}\left\{1 + \frac{1}{2}\frac{a}{b} - \frac{5}{8}\left(\frac{a}{b}\right)^2 + 0.421\left(\frac{a}{b}\right)^3\right\}$$

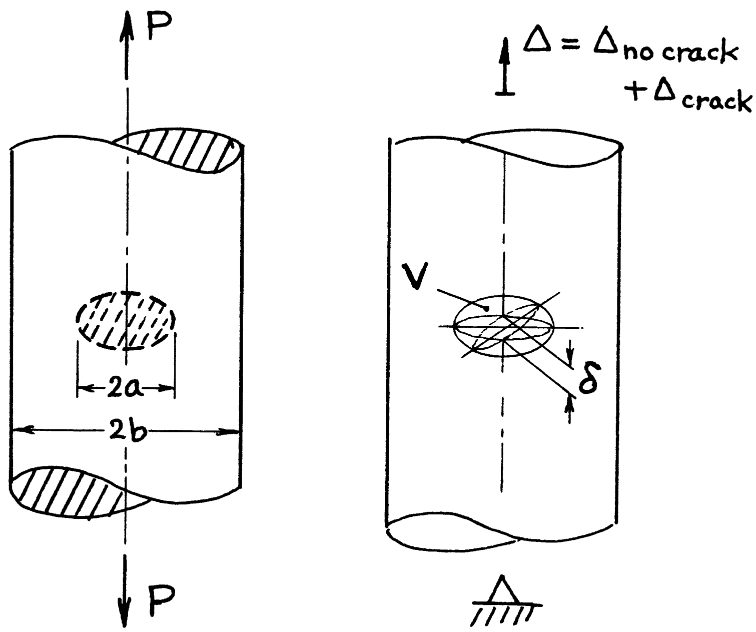
$$K_{II} = K_{III} = 0$$





Method: Asymptotic Approximation
Accuracy: Better than 1%
Reference: **Benthem 1972**

NOTE: For an alternate solution and displacements, see **page 27.4a**.



$$\sigma = \frac{P}{\pi b^2}$$

$$K_I = \frac{2}{\pi}\sigma\sqrt{\pi a} \cdot F\left(\frac{a}{b}\right)$$

$$F\left(\frac{a}{b}\right) = \frac{1 - \frac{1}{2}\frac{a}{b} + .148\left(\frac{a}{b}\right)^3}{\sqrt{1 - \frac{a}{b}}}$$

Volume of Crack:

$$V = \frac{16\left(1 - \nu^2\right)}{3E}\sigma a^3 \cdot G\left(\frac{a}{b}\right)$$

$$G\left(\frac{a}{b}\right) = \frac{1}{\left(a/b\right)^3}\left[1.260\,\ell n\left(\frac{1}{1 - \frac{a}{b}}\right) - 1.260\frac{a}{b} - .630\left(\frac{a}{b}\right)^2 + .580\left(\frac{a}{b}\right)^3 - .315\left(\frac{a}{b}\right)^4 - .102\left(\frac{a}{b}\right)^5 \right.$$
$$\left. + .063\left(\frac{a}{b}\right)^6 - .0093\left(\frac{a}{b}\right)^7 - .0081\left(\frac{a}{b}\right)^8\right]$$

Additional Elongation at Infinity due to Crack:

$$\Delta_{crack} = V\Big/\left(\pi b^2\right)$$

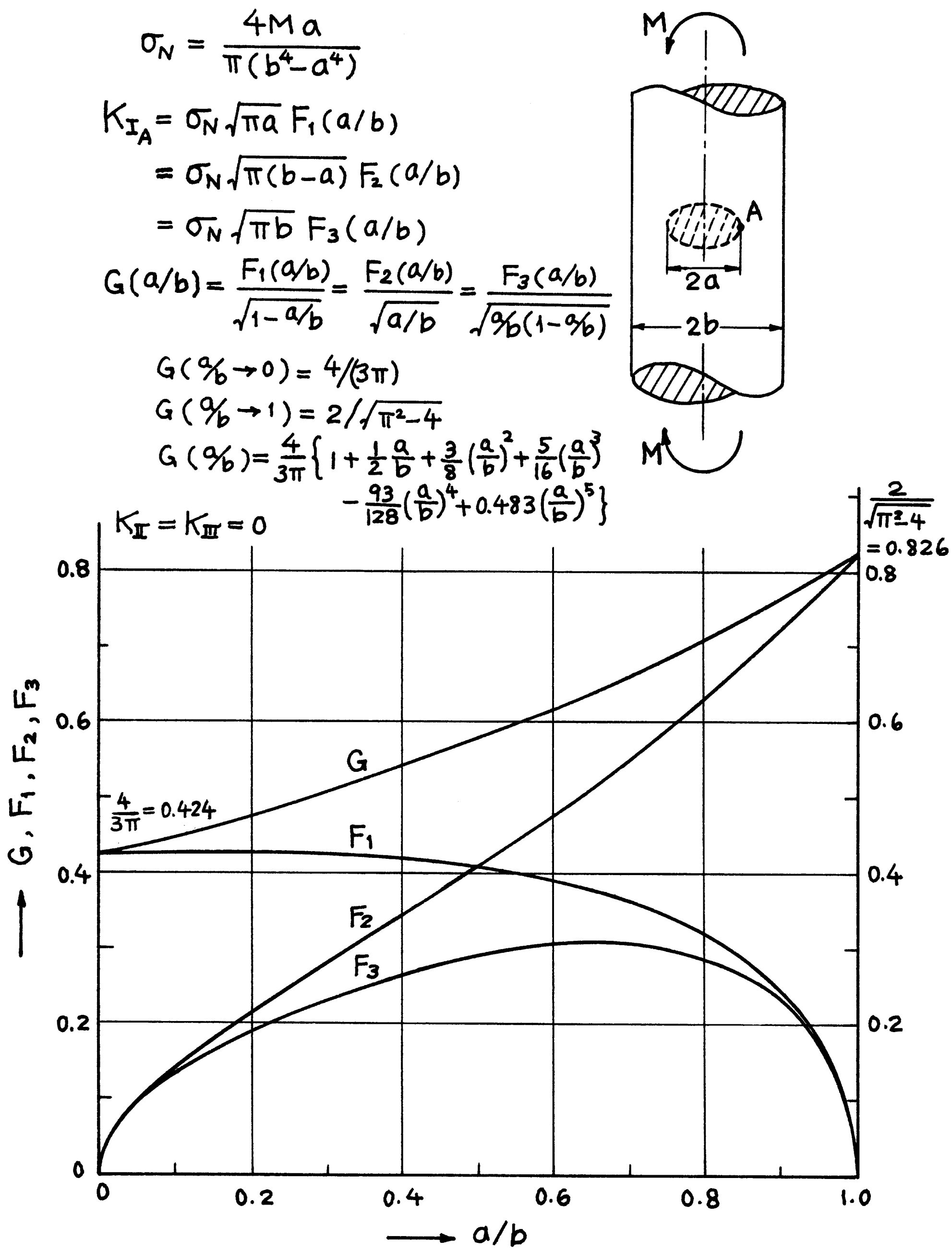
Crack Opening at Center:

$$\delta = \frac{8\left(1-\nu^2\right)}{\pi E}\sigma a \cdot H\left(\frac{a}{b}\right)$$

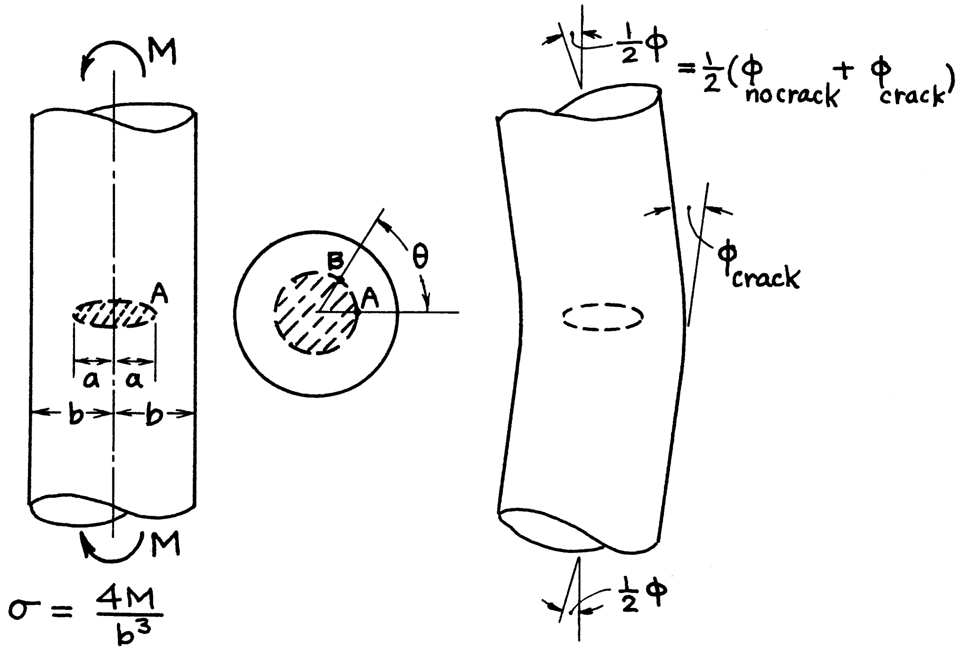
$$H\left(\frac{a}{b}\right) = \frac{1}{a/b}\ell n\left(\frac{1}{1-a/b}\right)\left[1-\frac{1}{2}\frac{a}{b}+.340\left(\frac{a}{b}\right)^{3.5}\right]$$

Method: K_I, δ Integral Transform; V, Δ Paris' Equation (see **Appendix B**)
Accuracy: K_I, δ 0.5%; V, Δ 1%
References: **Erdogan 1982; Tada 1985**

NOTE: Δ_{crack} is elongation at infinity when uniform pressure σ is applied on crack surfaces.



Method: Asymptotic Approximation
Accuracy: Better than 1%
Reference: **Benthem 1972**

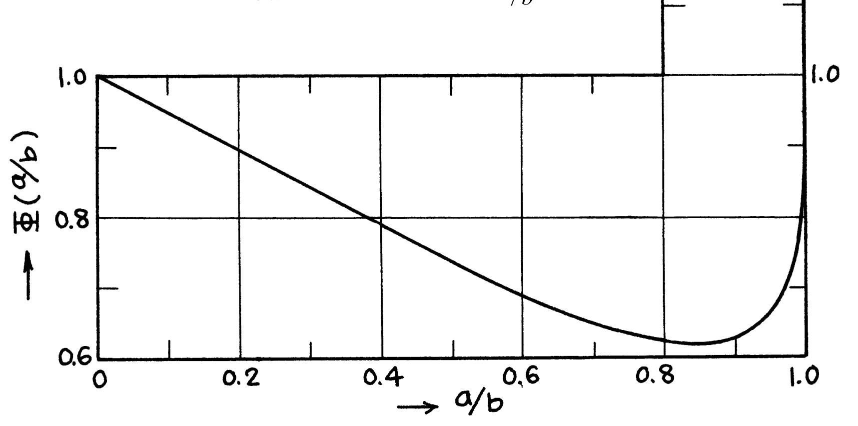


$$K_{IA} = \sigma\sqrt{\pi a}\,F\left({}^{a}\!/_{b}\right); \quad K_{IB} = K_{IA}\cos\theta$$

$$F\left({}^{a}\!/_{b}\right) = \frac{a}{b}\frac{\sqrt{1-{}^{a}\!/_{b}}}{\left\{1-\left({}^{a}\!/_{b}\right)^4\right\}}\cdot\frac{4}{3\pi}\left\{1+\frac{1}{2}\frac{a}{b}+\frac{3}{8}\left(\frac{a}{b}\right)^2+\frac{5}{16}\left(\frac{a}{b}\right)^3-\frac{93}{128}\left(\frac{a}{b}\right)^4+.483\left(\frac{a}{b}\right)^5\right\}$$

Additional Rotation at Infinity or Kink at Cracked Section due to Crack:

$$\phi_{crack} = \frac{1280(1-\nu^2)}{9\pi E}\sigma\left({}^{a}\!/_{b}\right)^4 \ell n\,\frac{1}{1-{}^{a}\!/_{b}}\cdot\Phi\left({}^{a}\!/_{b}\right)$$



Methods: K_I Asymptotic Interpolation; ϕ Paris' Equation (see **Appendix B**)
Accuracy: K_I 1%; ϕ 2%
References: **Benthem 1972; Tada 1985**

NOTE: Crack surface interference on compressive side is not considered.

$$\tau_N = \frac{2Ta}{\pi(b^4 - a^4)}$$

$$\begin{aligned} K_{III} &= \tau_N \sqrt{\pi a}\, F_1(a/b) \\ &= \tau_N \sqrt{\pi(b-a)}\, F_2(a/b) \\ &= \tau_N \sqrt{\pi b}\, F_3(a/b) \end{aligned}$$

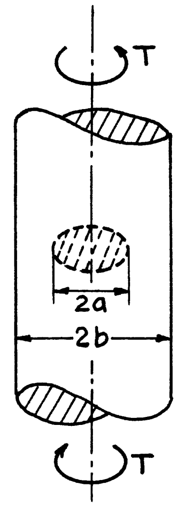
$$G(a/b) = \frac{F_1(a/b)}{\sqrt{1-a/b}} = \frac{F_2(a/b)}{\sqrt{a/b}} = \frac{F_3(a/b)}{\sqrt{a/b\,(1-a/b)}}$$

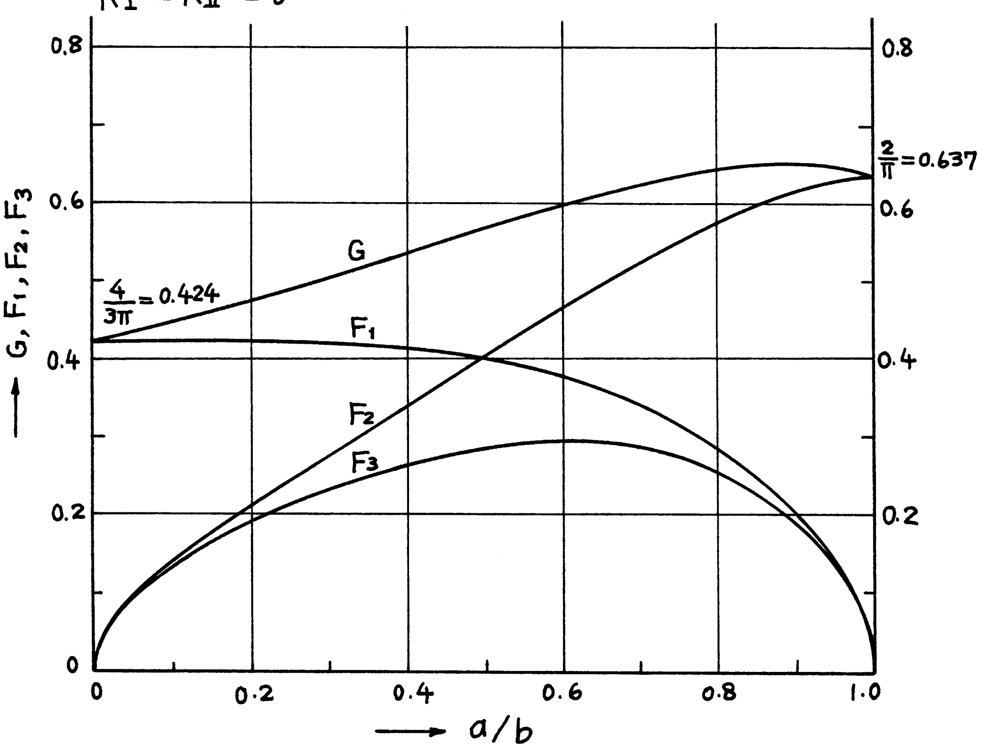
$$G(a/b \to 0) = 4/(3\pi)$$

$$G(a/b \to 1) = 2/\pi$$

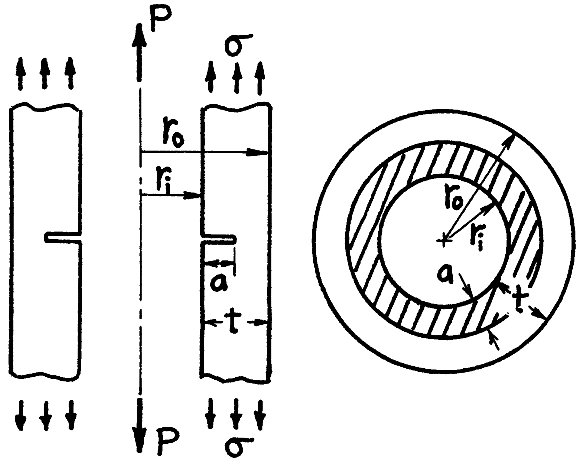
$$G(a/b) = \frac{4}{3\pi}\left\{1 + \frac{1}{2}\frac{a}{b} + \frac{3}{8}\left(\frac{a}{b}\right)^2 + \frac{5}{16}\left(\frac{a}{b}\right)^3 - \frac{93}{128}\left(\frac{a}{b}\right)^4 + 0.038\left(\frac{a}{b}\right)^5\right\}$$

$$K_I = K_{II} = 0$$



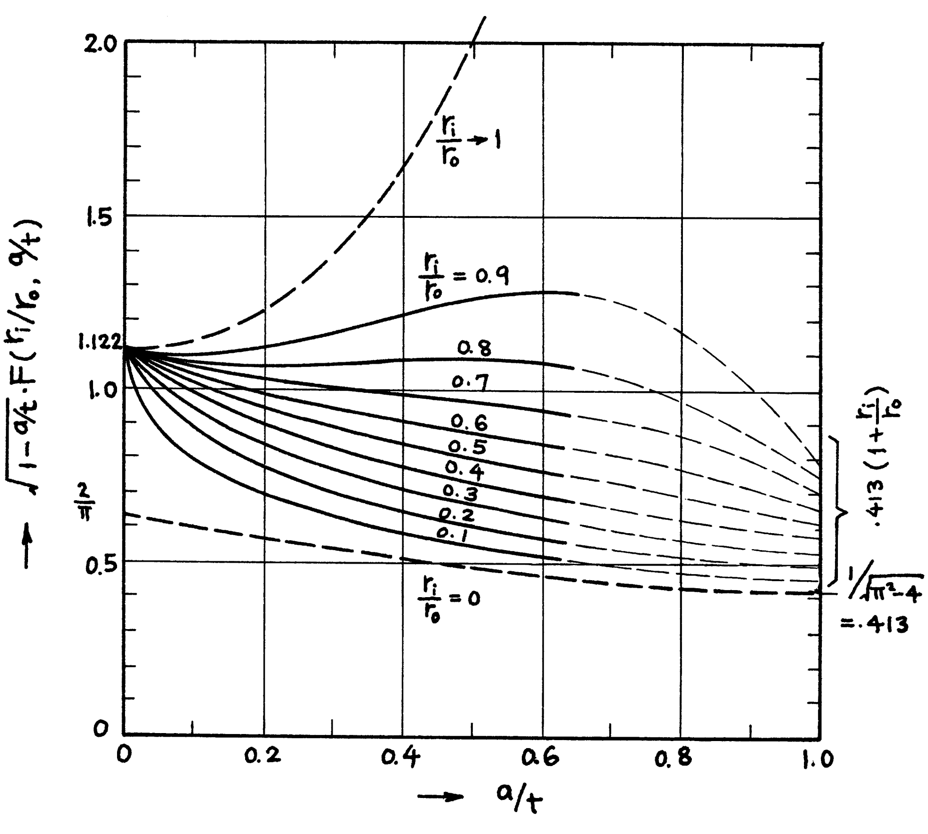


Method: Asymptotic Approximation
Accuracy: Better than 1%
Reference: **Benthem 1972**



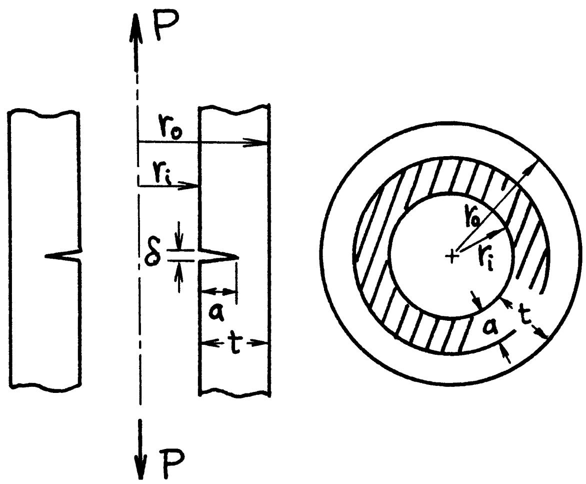
$$\sigma = \frac{P}{\pi\left(r_o^2 - r_i^2\right)}$$

$$K_I = \sigma\sqrt{\pi a} \cdot F\left(\frac{r_i}{r_o}, \frac{a}{t}\right)$$



Method: Integral Transform-Singular Integral Equation $(a/t \le 0.6)$, Interpolation $(a/t > 0.6)$
Accuracy: Solid curves $\left(0.1 \le r_i/r_o \le 0.9, a/t \le 0.6\right)$ are based on values with better than 1% accuracy; 2% for $a/t > 0.6$
References: **Erdogan 1982; Tada 1985**

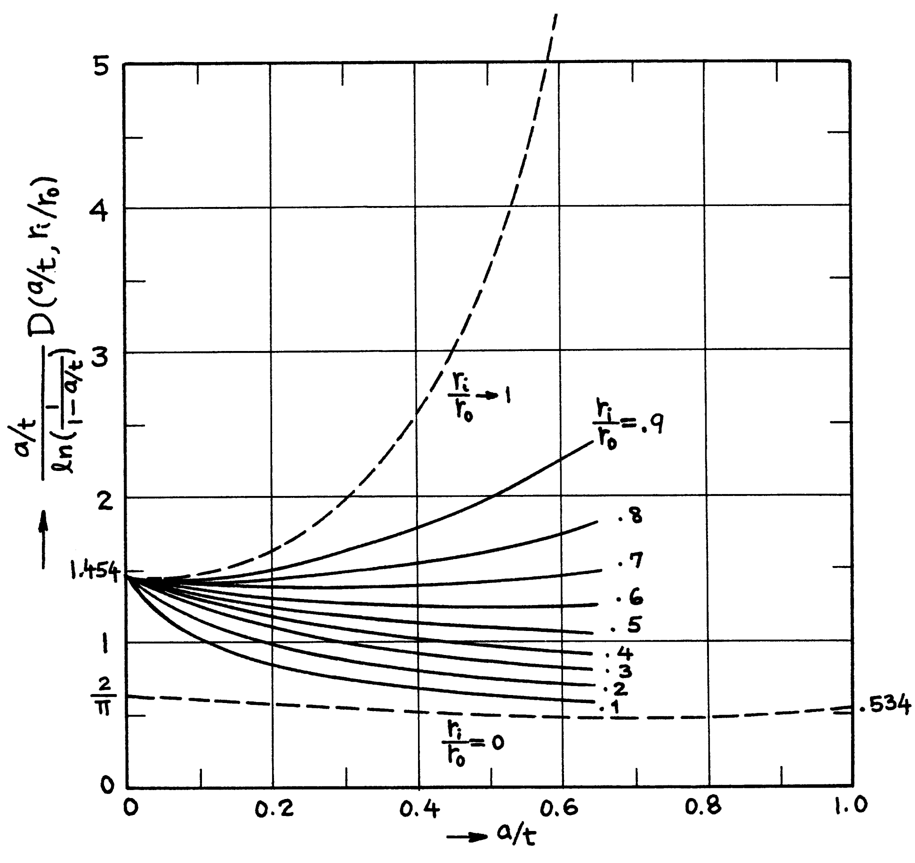
NOTE: For $r_i/r_o = 0$ (solid cylinders), see **page 27.4**, and for $r_i/r_o \to 1$, see **page 2.10**.



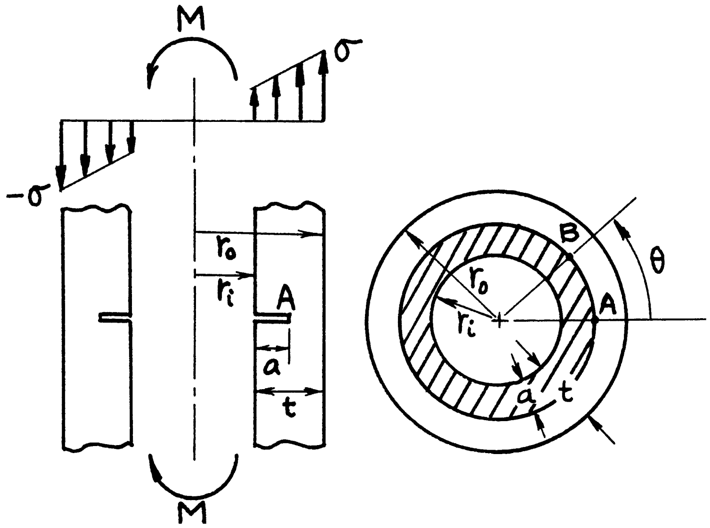
$$\sigma = \frac{P}{\pi\left(r_o^2 - r_i^2\right)}$$

Crack Opening at Edge:

$$\delta = \frac{4(1-\nu^2)}{E}\sigma a \cdot D\left(a/t, r_i/r_o\right)$$



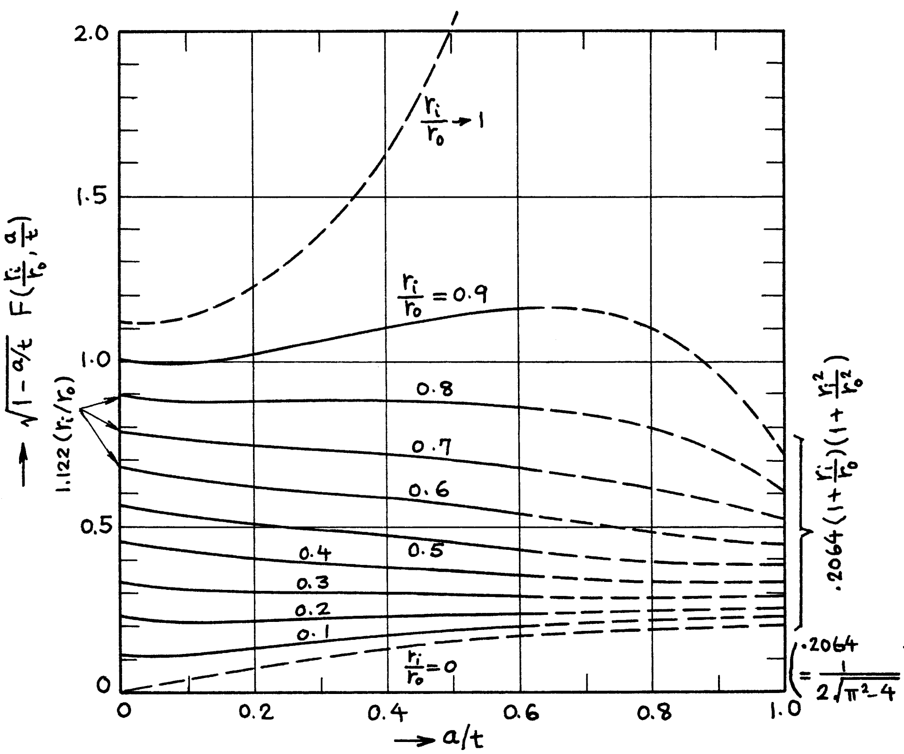
Method: Integral Transform-Integral Equations
Accuracy: Curves are based on values with better than 1% accuracy.
References: **Erdogan 1982; Tada 1985**



$$\sigma = \frac{4Mr_o}{\pi\left(r_o^4 - r_i^4\right)}$$

$$K_{IA} = \sigma\sqrt{\pi a}\,F\left(\frac{r_i}{r_o}, \frac{a}{t}\right)$$

$$K_{IB} = K_{IA}\cos\theta$$

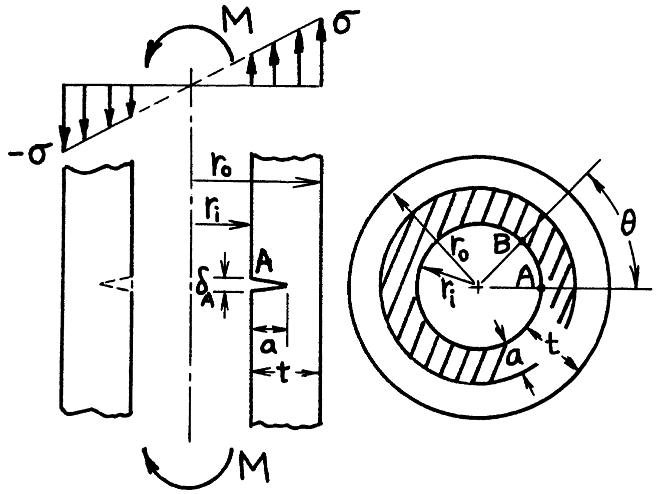


Methods: Integral Transform-Integral Equations $(a/t \leq 0.6)$, Interpolation $(a/t > 0.6)$.

Accuracy: Solid curves $\left(0.1 \leq r_i/r_o < 0.9; a/t \leq 0.6\right)$ are based on values with better than 1% accuracy; 2% for $a/t > 0.6$.

References: **Erdogan 1982; Tada 1985**.

NOTE: Crack surface interference on compression side is not considered. For $r_i/r_o = 0$ (solid cylinder), see **page 27.5**; for $r_i/r_o \to 1$, see **page 2.10**.

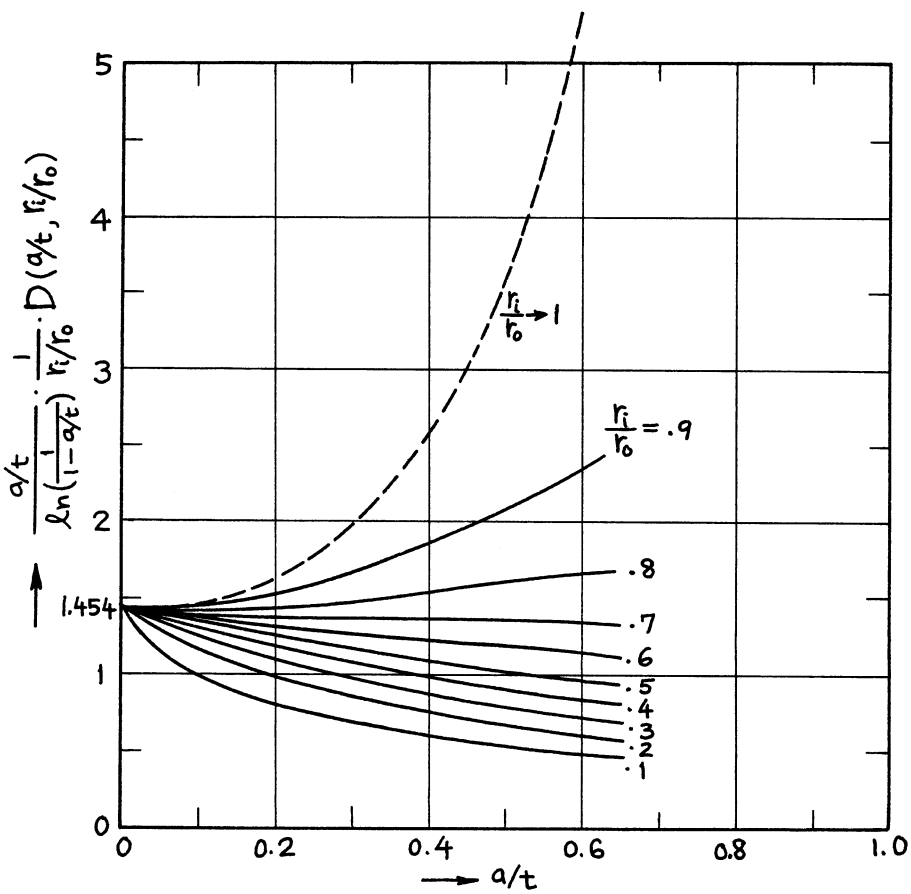


$$\sigma = \frac{4Mr_o}{\pi\left(r_o^4 - r_i^4\right)}$$

Crack Opening at Edge:

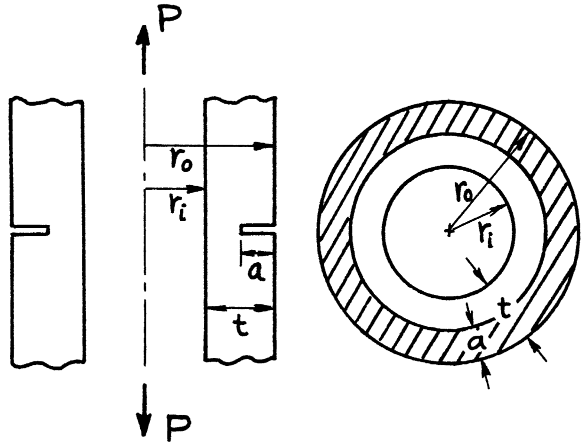
$$\delta_A = \frac{4\left(1-\nu^2\right)}{E}\sigma a \cdot D\left(a/t, r_i/r_o\right)$$

$$\delta_B = \delta_A \cos\theta$$



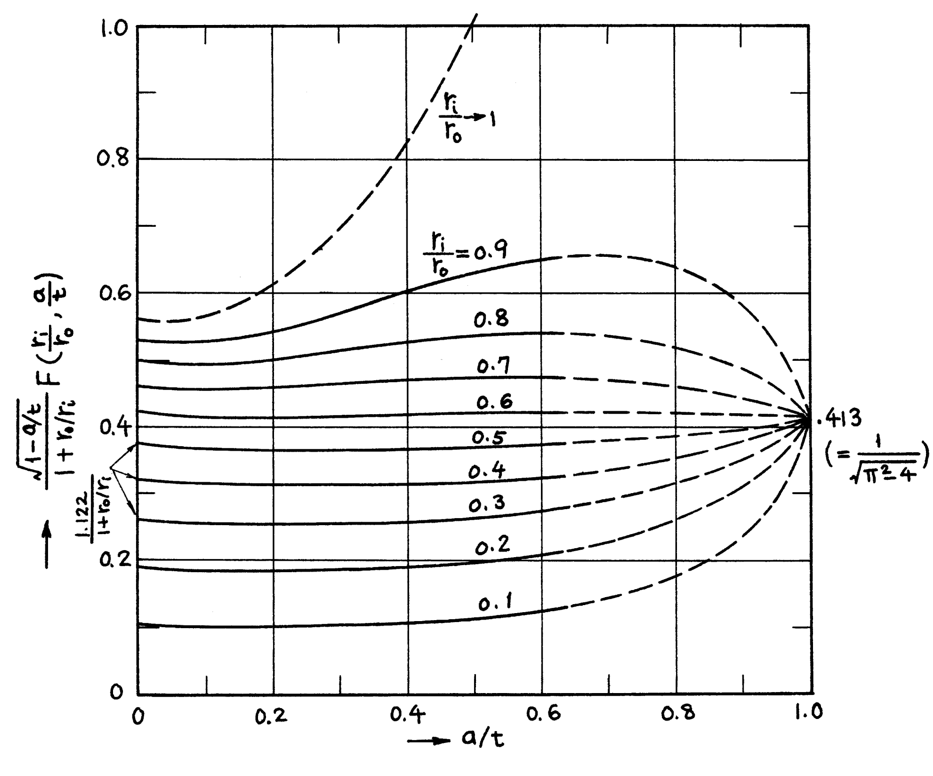
Method: Integral Transform - Integral Equations
Accuracy: Curves are based on values with better than 1% accuracy.
References: **Erdogan 1982; Tada 1985**

NOTE: Crack surface interference on compressive side is not considered.



$$\sigma = \frac{P}{\pi\left(r_o^2 - r_i^2\right)}$$

$$K_I = \sigma\sqrt{\pi a}\, F\left(\frac{r_i}{r_o}, \frac{a}{t}\right)$$

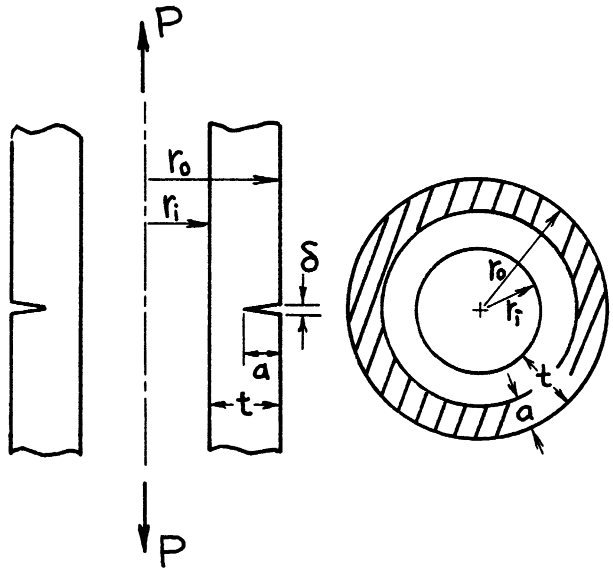


Methods: Integral Transform - Integral Equations $(a/t \leq 0.6)$, Interpolation $(a/t > 0.6)$

Accuracy: Solid curves $(0.1 \leq r_i/r_o \leq 0.9; a/t \leq 0.6)$ are based on values with better than 1% accuracy; 2% for $a/t > 0.6$.

References: **Erdogan 1982; Tada 1985**

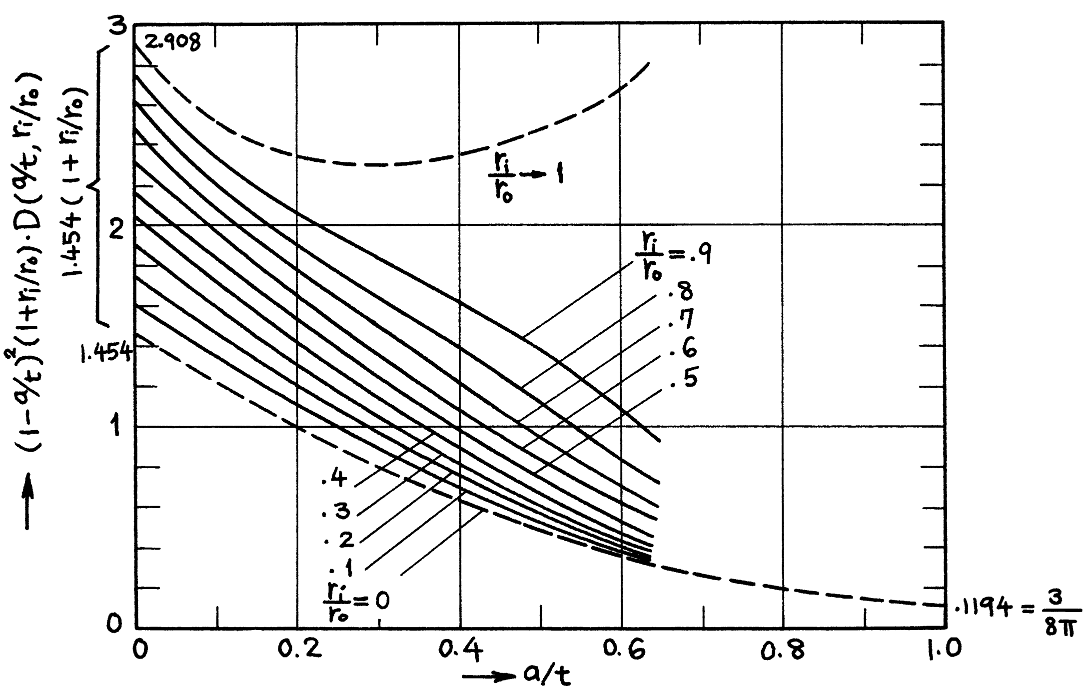
NOTE: For $r_i/r_o = 0$ (solid cylinder), see **page 27.1**; for $r_i/r_o \to 1$, see **page 2.10**.



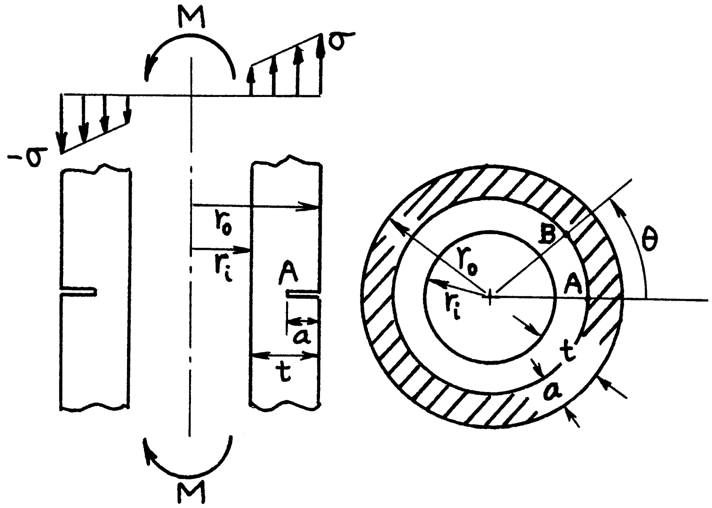
$$\sigma = \frac{P}{\pi\left(r_o^2 - r_i^2\right)}$$

Crack Opening at Edge:

$$\delta = \frac{4(1-\nu^2)}{E}\sigma a \cdot D\left({}^{a}\!/_{t}, {}^{r_i}\!/_{r_o}\right)$$



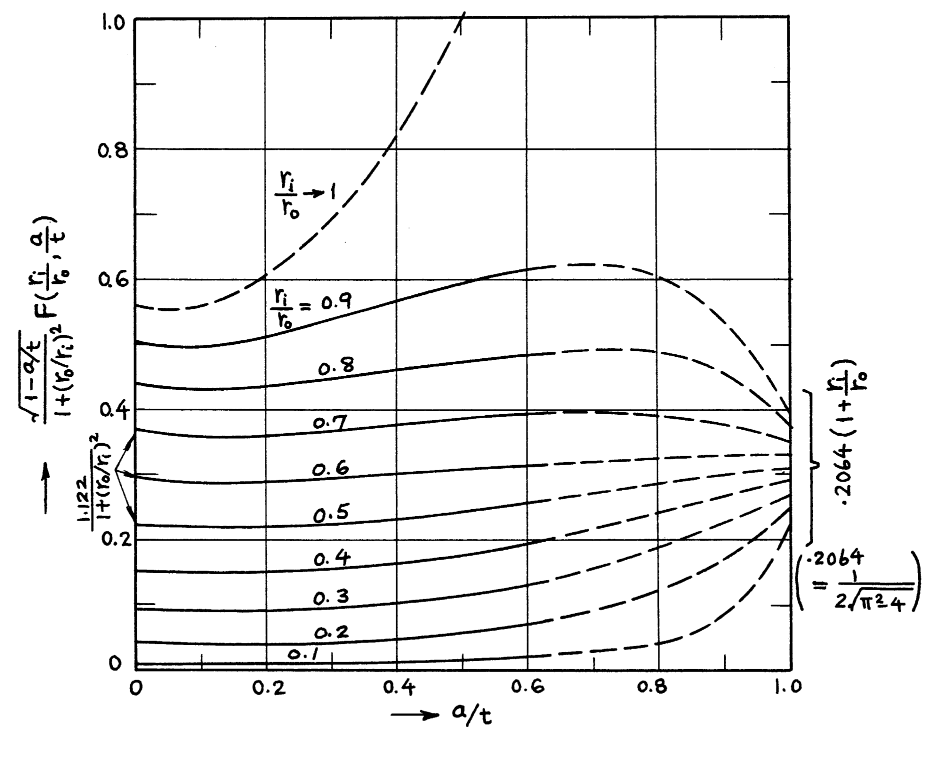
Method: Integral Transform - Integral Equations
Accuracy: Curves are based on values with better than 1% accuracy.
References: **Erdogan 1982; Tada 1985**



$$\sigma = \frac{4Mr_o}{\pi\left(r_o^4 - r_i^4\right)}$$

$$K_{IA} = \sigma\sqrt{\pi a}\,F\left(\frac{r_i}{r_o}, \frac{a}{t}\right)$$

$$K_{IB} = K_{IA}\cos\theta$$

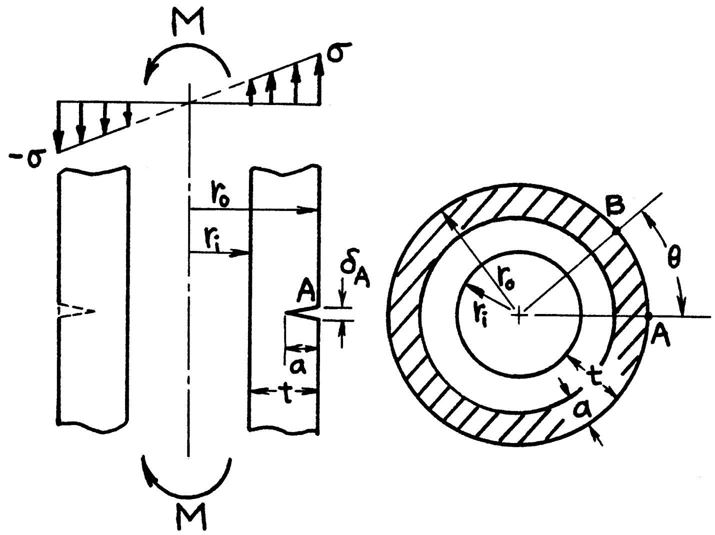


Methods: Integral Transform - Integral Equations $(a/t \leq 0.6)$, Interpolation $(a/t > 0.6)$

Accuracy: Solid curves $(0.1 \leq r_i/r_o \leq 0.9; a/t \leq 0.6)$ are based on values with better than 1% accuracy; 2% for $a/t > 0.6$.

References: **Erdogan 1982; Tada 1985**

NOTE: Crack surface interference on compression side is not considered. For $r_i/r_o = 0$ (solid cylinder), see **page 27.2**; for $r_i/r_o \to 1$, see **page 2.10**.

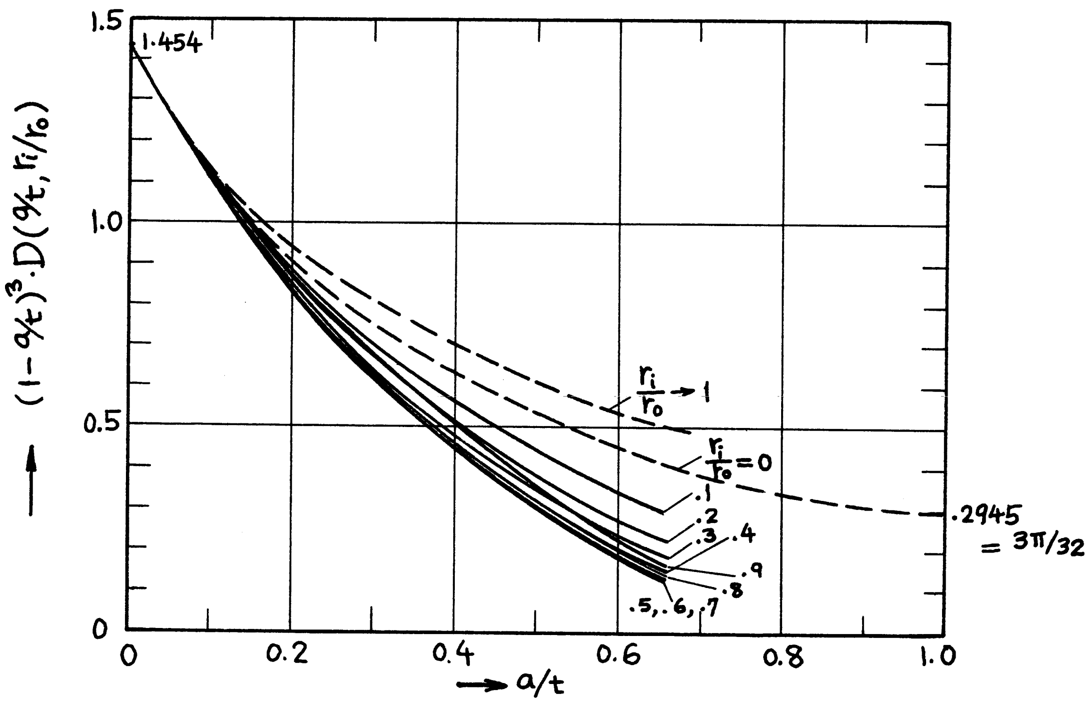


$$\sigma = \frac{4Mr_o}{\pi\left(r_o^4 - r_i^4\right)}$$

Crack Opening at Edge:

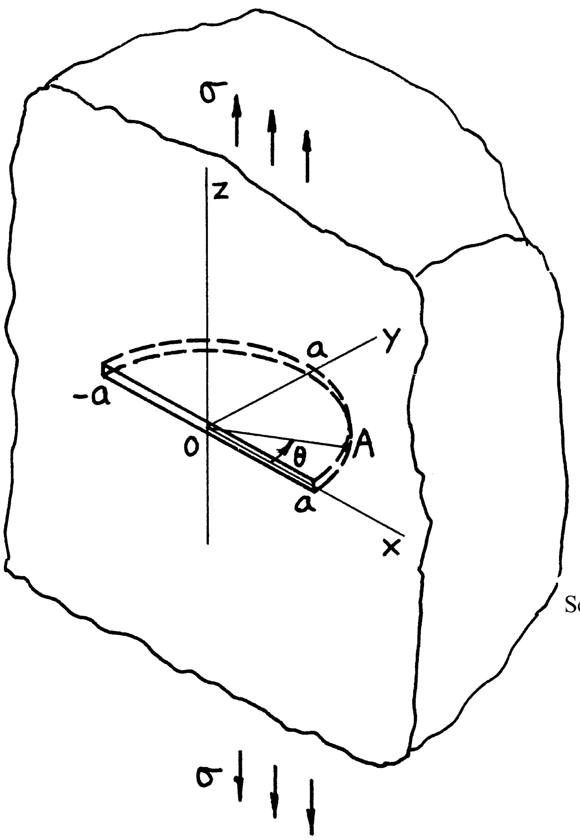
$$\delta_A = \frac{4(1-\nu^2)}{E}\sigma a \cdot D\left({}^{a}\!/_{t}, {}^{r_i}\!/_{r_o}\right)$$

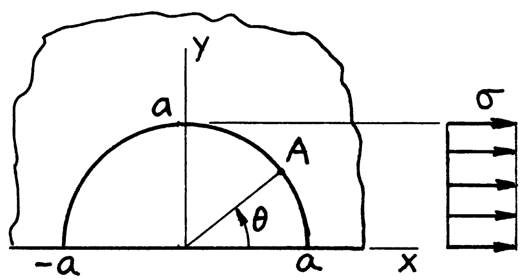
$$\delta_B = \delta_A \cos\theta$$



Method: Integral Transform - Integral Equations
Accuracy: Curves are based on values with better than 1% accuracy.
References: **Erdogan 1982; Tada 1985**

NOTE: Crack surface interference on compressive side is not considered.





Semi-infinite body $(y \geq 0)$ $(y = 0$: Free surface)

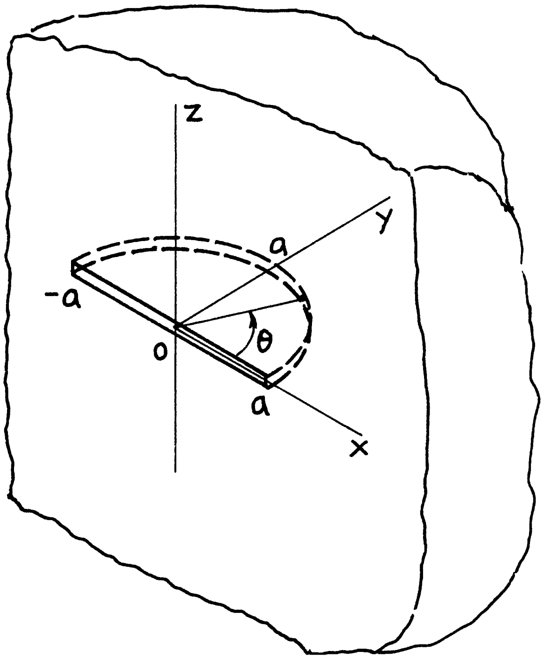
$$K_{IA} = \frac{2}{\pi}\sigma\sqrt{\pi a}\,F(\theta)$$

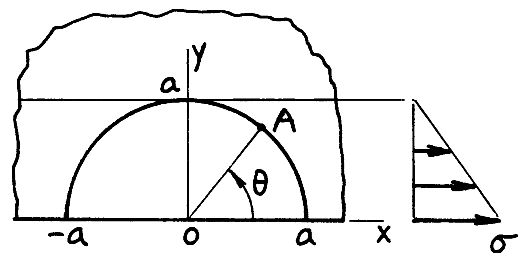
$$F(\theta) = 1.211 - .186\sqrt{\sin\theta} \quad \left(10^{\circ} < \theta < 170^{\circ}\right)$$

Methods: Alternating Method (Smith, Hartranft), Finite Element Method (Tracey, Raju); $F(\theta)$ is based on Smith's result (Merkle)

Accuracy: 2%

References: **Smith 1967; Hartranft 1973; Tracey 1973; Merkle 1973; Raju 1979**





$$\sigma_z(x,y,0) = \sigma\left(1-\frac{y}{a}\right)$$

Semi-infinite body $(y \geq 0)$ $(y = 0$: Free surface)

$$K_{IA} = \frac{2}{\pi}\sigma\sqrt{\pi a}\cdot F(\theta)$$

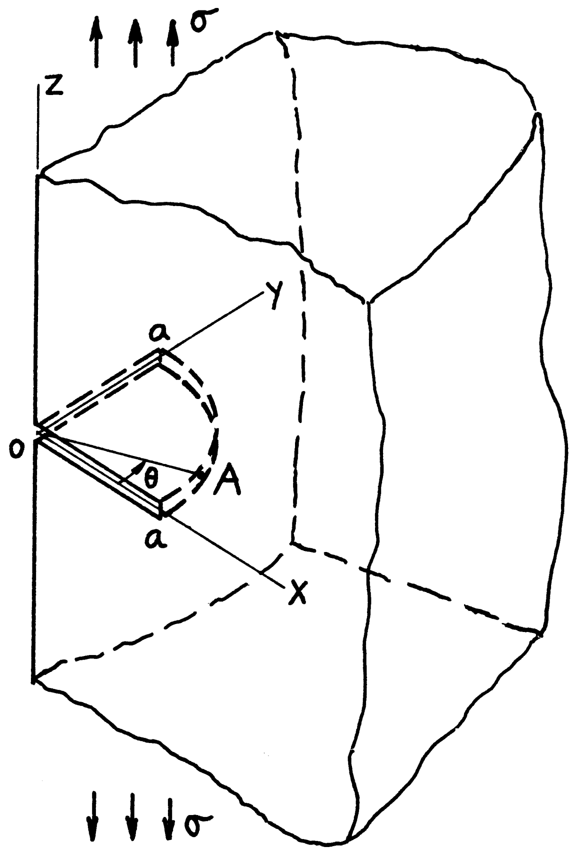
$$or \quad = \frac{2}{\pi}\sigma\sqrt{\pi a}\left\{\frac{5}{6}-\frac{4}{3}(\sin\theta)^{3/2}\left(\sqrt{1+\sin\theta}-\sqrt{\sin\theta}\right)\right\}\cdot G(\theta)$$

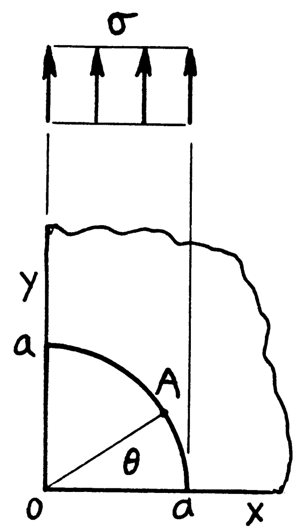
$$F(\theta) = 1.031 - .186\sqrt{\sin\theta} - .54\sin\theta$$

$$(10^\circ < \theta < 170^\circ)$$

$$G(\theta) = 1.17 - .31\sin\theta + .23(\sin\theta)^2$$

(See **page 24.16**)
Methods: Approximations ($F(\theta)$, Merkle; $G(\theta)$, Tada) of Numerical Results by Alternating Method (Smith)
Accuracy: 3%
References: **Smith 1967; Merkle 1973; Tada 1975**





($x = 0$ and $y = 0$: Free surfaces)

$$K_{IA} = \frac{2}{\pi}\sigma\sqrt{\pi a}\,F_Q(\theta)$$

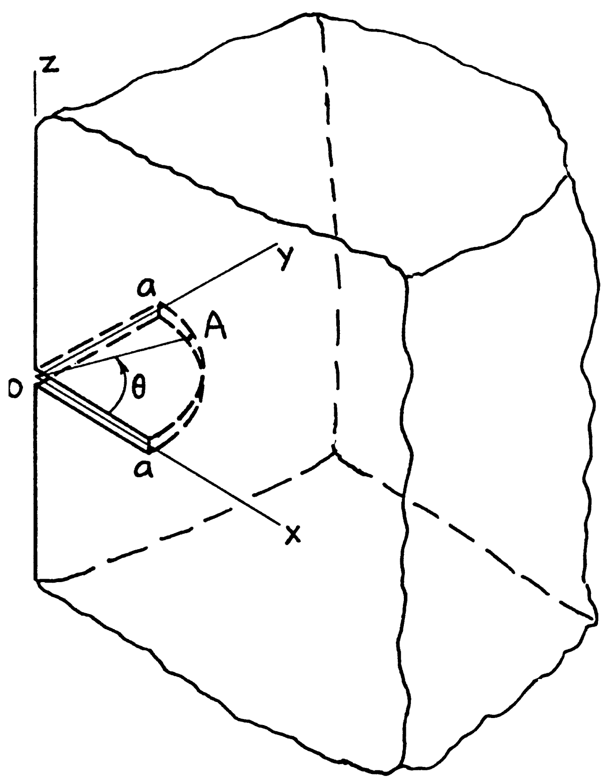
$$F_Q(\theta) = F(\theta)\cdot F\left(\frac{\pi}{2} - \theta\right)$$

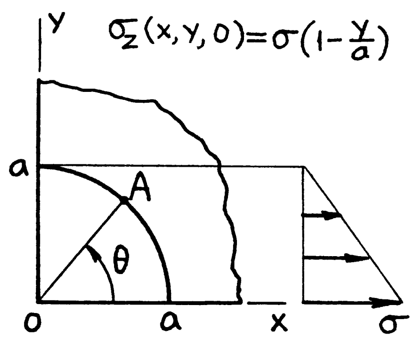
$$F(\theta) = 1.211 - .186\sqrt{\sin\theta} \qquad \left(10^\circ < \theta < 80^\circ\right)$$

$$or \quad F_Q(\theta) = \left(1.211 - .186\sqrt{\sin\theta}\right)\left(1.211 - .186\sqrt{\cos\theta}\right)$$

Method: Alternating Method (Kobayashi), Finite Element Method (Tracey, Newman)
Accuracy: 3%
References: **Kobayashi 1976; Tracey 1973; Newman 1981a; Tada 1985**

NOTE: See **page 28.1**. $F(\theta)$ is the free surface correction for a half-circular surface crack. Total correction for a quarter-circular corner crack is approximately equal to product of surface corrections for a half-circular crack.





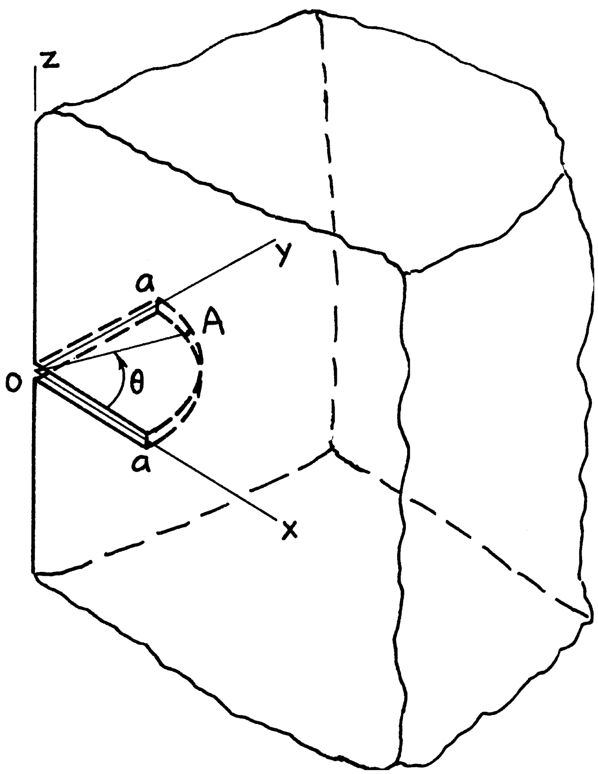
$(x = 0$ and $y = 0$: Free surfaces)

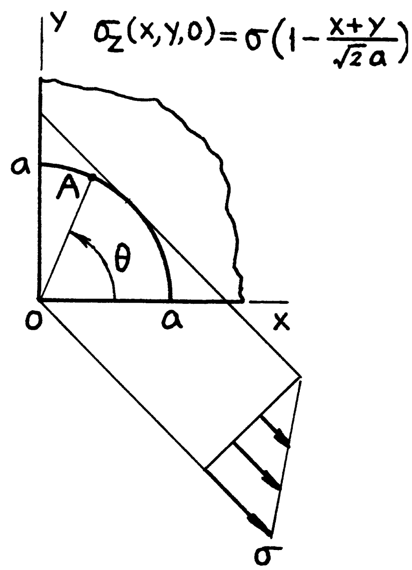
$$K_{IA} = \frac{2}{\pi}\sigma\sqrt{\pi a}\cdot F(\theta)$$

$$or \quad = \frac{2}{\pi}\sigma\sqrt{\pi a}\left\{\frac{5}{6} - \frac{4}{3}(\sin\theta)^{3/2}\left(\sqrt{1+\sin\theta} - \sqrt{\sin\theta}\right)\right\}\cdot G(\theta)$$

$$\left.\begin{aligned} F(\theta) &= 1 - .72\sin\theta + .11(\sin\theta)^2 \\ G(\theta) &= 1.22 - .56\sin\theta + .70(\sin\theta)^2 \end{aligned}\right. \qquad (10^\circ < \theta < 80^\circ)$$

(See **page 24.16**)
Method: Approximation of Numerical Results by Alternating Method (Kobayashi)
Accuracy: 3%
References: **Kobayashi 1976; Tada 1975**





($x = 0$ and $y = 0$: Free surfaces)

$$K_{IA} = \tfrac{2}{\pi}\sigma\sqrt{\pi a} \cdot F(\theta)$$

$$or \qquad = \tfrac{2}{\pi}\sigma\sqrt{\pi a}\left\{\left(1 + \frac{1}{\sqrt{2}}\right) - \frac{2\sqrt{2}}{3}\left[(\cos\theta)^{3/2}\sqrt{1+\cos\theta} + (\sin\theta)^{3/2}\sqrt{1+\sin\theta}\right]\right\} \cdot G(\theta)$$

$$F(\theta) = .311 + .154\frac{\left(1 - \overline{\theta}^2\right)^2}{1 + 4\overline{\theta}^2}; \overline{\theta} = \frac{\theta}{\pi/4}$$

$$(10^{\circ} < \theta < 80^{\circ})$$

$$G(\theta) = 1.245 + .04(\sin 2\theta)^2$$

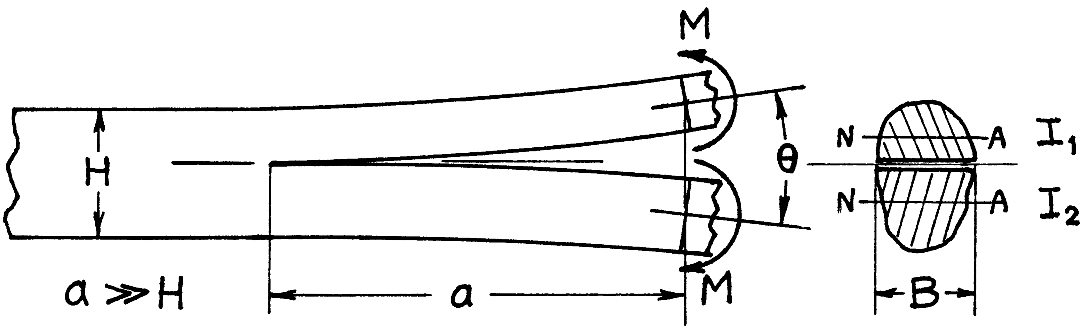
(See **page 24.17**)
Method: Superposition of **page 28.4** (and also **page 28.3**)
Accuracy: $F(\theta)$ 4%; $G(\theta)$ 3%
Reference: **Tada 1985**

PART

V

CRACK(S) IN A ROD OR A PLATE BY ENERGY RATE ANALYSIS

- ❏ (Bending, Shearing, and Tension/Compression)



$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II}$$

$$= \frac{M^2}{2EIB}$$

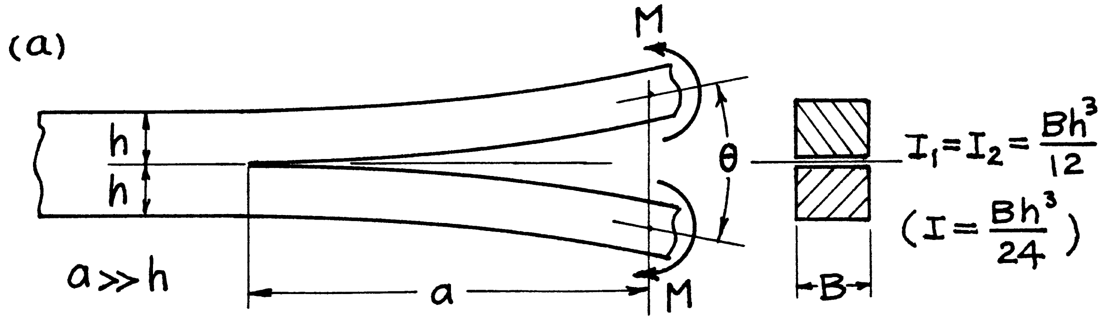
or

$$= \frac{EI\theta^2}{2a^2B}$$

where $$\frac{1}{I} = \frac{1}{I_1} + \frac{1}{I_2}$$

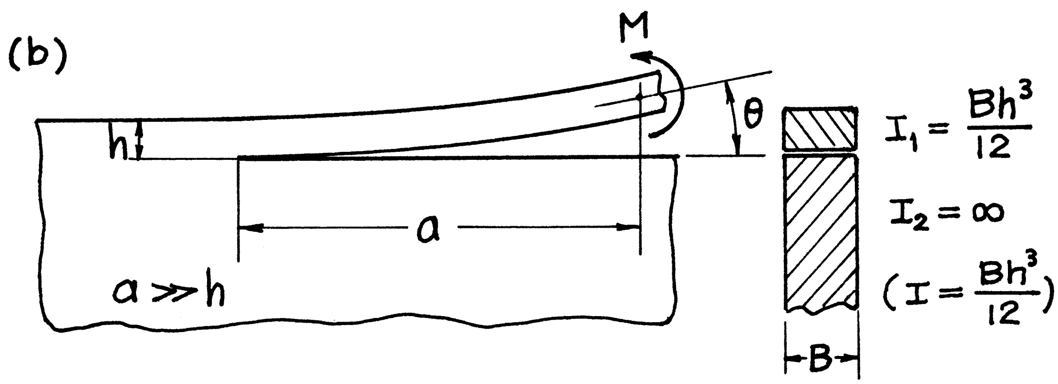
$I_1, I_2 =$ moment of inertia of each cross section about its neutral axis

Method: Energy Balance
Accuracy: Approximation by Simple Beam Theory
References: **Tada 1974, 2000**



$$\mathcal{G} = \mathcal{G}_I = \frac{12\left(M/B\right)^2}{E'h^3} \overset{\text{or}}{=} \frac{E'h^3}{48\,a^2}\theta^2$$

$$K_I = \frac{2\sqrt{3}\left(M/B\right)}{h^{3/2}} \overset{\text{or}}{=} \frac{E'h^{3/2}}{4\sqrt{3}\,a}\theta$$



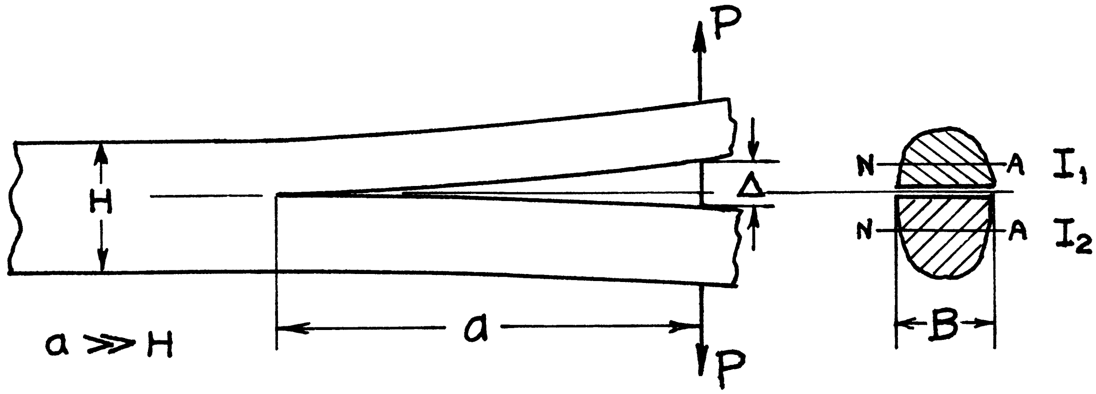
$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II} = \frac{6\left(M/B\right)^2}{E'h^3} \overset{\text{or}}{=} \frac{E'h^3}{24\,a^2}\theta^2$$

$$\mathcal{G}_I = 0.6222\mathcal{G}, \quad \mathcal{G}_{II} = 0.3778\mathcal{G}$$

$$K_I = 0.7888\frac{\sqrt{6}\left(M/B\right)}{h^{3/2}} \overset{\text{or}}{=} 0.7888\frac{E'h^{3/2}}{2\sqrt{6}\,a}\theta$$

$$K_{II} = -0.6147\frac{\sqrt{6}\left(M/B\right)}{h^{3/2}} \overset{\text{or}}{=} -0.6147\frac{E'h^{3/2}}{2\sqrt{6}\,a}\theta$$

(a) Special Case of **29.1**
(b) Special Case of **29.1** and **29.8**; see **Hutchinson 1992**



$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II}$$

$$= \frac{a^2}{2EIB} P^2$$

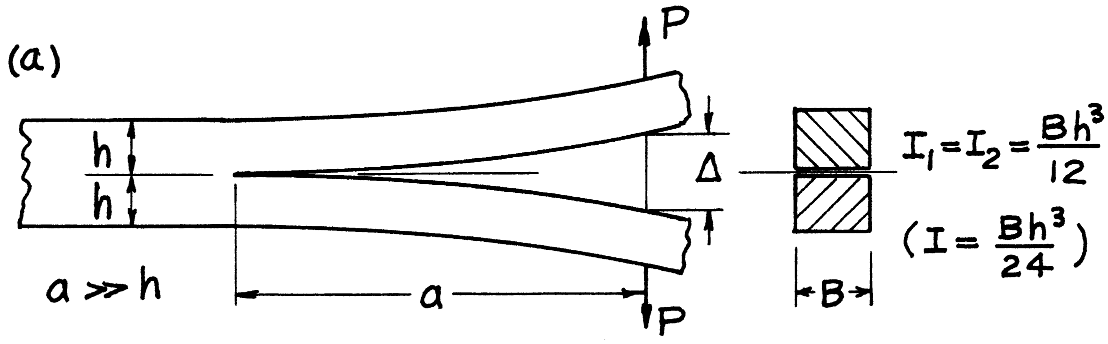
or

$$= \frac{9EI}{2a^4 B} \Delta^2$$

where $$\frac{1}{I} = \frac{1}{I_1} + \frac{1}{I_2}$$

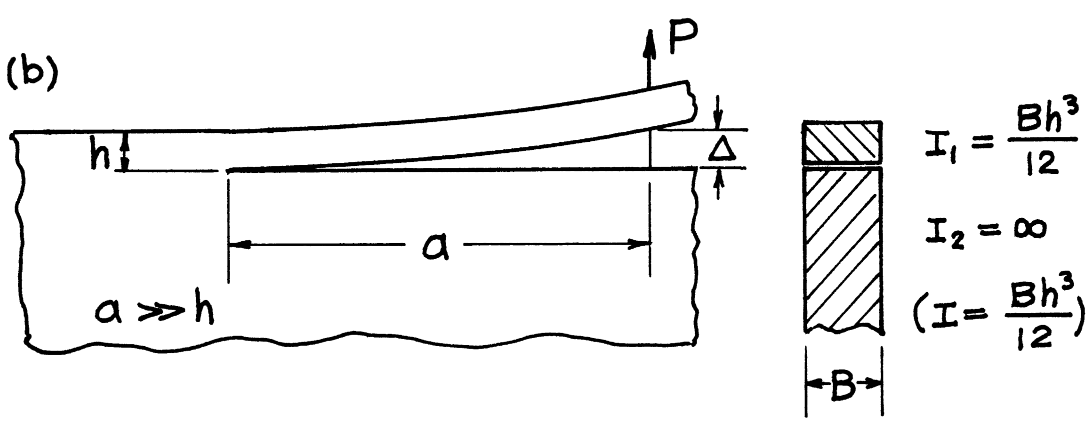
$I_1, I_2 =$ moment of inertia of each cross section about its neutral axis

Method: Energy Balance
Accuracy: Approximation by Simple Beam Theory
References: **Tada 1974, 2000**



$$\mathcal{G} = \mathcal{G}_I = \frac{12\,a^2}{E'h^3}\left(\frac{P}{B}\right)^2 \overset{\text{or}}{=} \frac{3\,E'h^3}{16\,a^4}\Delta^2$$

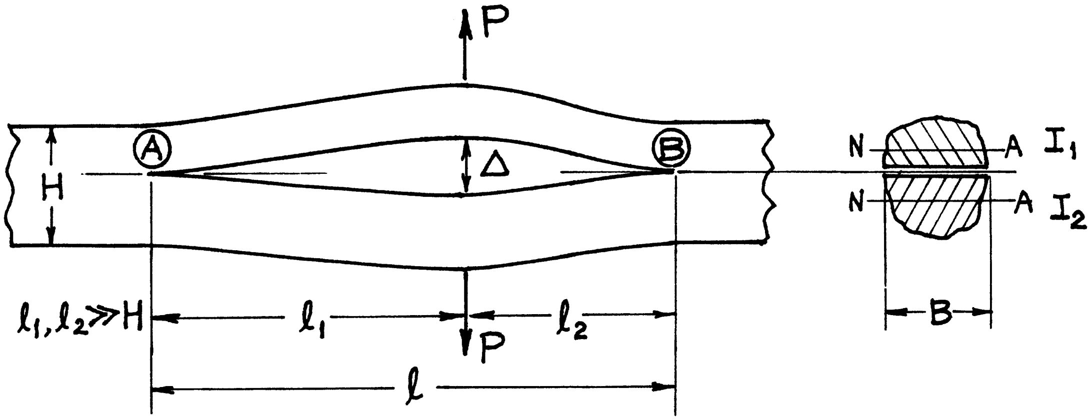
$$K_I = \frac{2\sqrt{3}\,a}{h^{3/2}}\left(\frac{P}{B}\right) \overset{\text{or}}{=} \frac{\sqrt{3}\,E'h^{3/2}}{4\,a^2}\Delta$$



$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II}$$

$$= \frac{6\,a^2}{E'h^3}\left(\frac{P}{B}\right)^2 \overset{\text{or}}{=} \frac{3\,E'h^3}{8\,a^4}\Delta^2$$

Special cases of **29.3**



$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II}$$

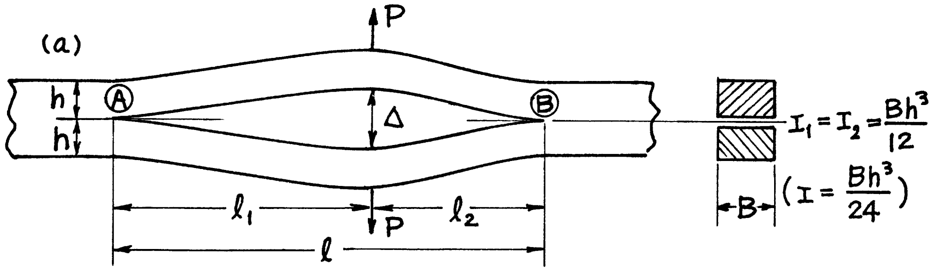
$$\mathcal{G}_{Ⓐ} = \frac{1}{2EIB} \frac{\ell_1^2 \ell_2^4}{\ell^4} P^2 \quad \overset{\text{or}}{=} \quad \frac{9EI}{2B} \frac{\ell^2}{\ell_1^4 \ell_2^2} \Delta^2$$

$$\mathcal{G}_{Ⓑ} = \frac{1}{2EIB} \frac{\ell_1^4 \ell_2^2}{\ell^4} P^2 \quad \overset{\text{or}}{=} \quad \frac{9EI}{2B} \frac{\ell^2}{\ell_1^2 \ell_2^4} \Delta^2$$

$$\text{where} \quad \frac{1}{I} = \frac{1}{I_1} + \frac{1}{I_2}$$

$I_1, I_2 =$ moment of inertia of each cross section about its neutral axis

Method: Energy Balance
Accuracy: Approximation by Simple Beam Theory
References: **Tada 1974, 2000**



$\ell_1, \ell_2 \gg h$

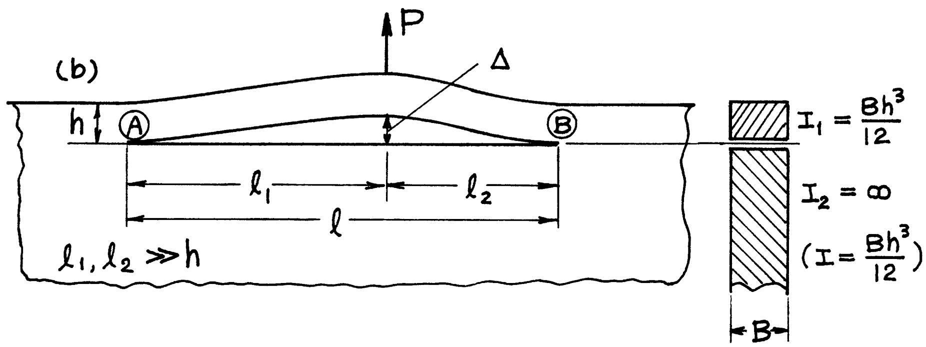
$$\mathcal{G} = \mathcal{G}_I$$

$$\mathcal{G}_{I\text{Ⓐ}} = \frac{12}{E'h^3}\frac{\ell_1^2\,\ell_2^4}{\ell^4}\left(\frac{P}{B}\right)^2 \overset{\text{or}}{=} \frac{3\,E'h^3}{16}\frac{\ell^2}{\ell_1^4\,\ell_2^2}\Delta^2$$

$$\mathcal{G}_{I\text{Ⓑ}} = \frac{12}{E'h^3}\frac{\ell_1^4\,\ell_2^2}{\ell^4}\left(\frac{P}{B}\right)^2 \overset{\text{or}}{=} \frac{3\,E'h^3}{16}\frac{\ell^2}{\ell_1^2\,\ell_2^4}\Delta^2$$

$$K_{I\text{Ⓐ}} = \frac{2\sqrt{3}}{h^{3/2}}\frac{\ell_1\,\ell_2^2}{\ell^2}\left(\frac{P}{B}\right) \overset{\text{or}}{=} \frac{\sqrt{3}\,E'h^{3/2}}{4}\frac{\ell}{\ell_1^2\,\ell_2}\Delta$$

$$K_{I\text{Ⓑ}} = \frac{2\sqrt{3}}{h^{3/2}}\frac{\ell_1^2\,\ell_2}{\ell^2}\left(\frac{P}{B}\right) \overset{\text{or}}{=} \frac{\sqrt{3}\,E'h^{3/2}}{4}\frac{\ell}{\ell_1\,\ell_2^2}\Delta$$

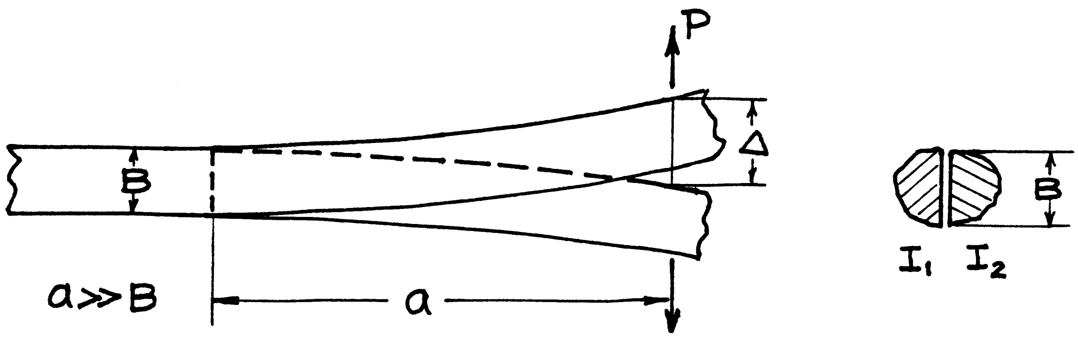


$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II}$$

$$\mathcal{G}_{\text{Ⓐ}} = \frac{6}{E'h^3}\frac{\ell_1^2\,\ell_2^4}{\ell^4}\left(\frac{P}{B}\right)^2 \overset{\text{or}}{=} \frac{3\,E'h^3}{8}\frac{\ell^2}{\ell_1^4\,\ell_2^2}\Delta^2$$

$$\mathcal{G}_{\text{Ⓑ}} = \frac{6}{E'h^3}\frac{\ell_1^4\,\ell_2^2}{\ell^4}\left(\frac{P}{B}\right)^2 \overset{\text{or}}{=} \frac{3\,E'h^3}{8}\frac{\ell^2}{\ell_1^2\,\ell_2^4}\Delta^2$$

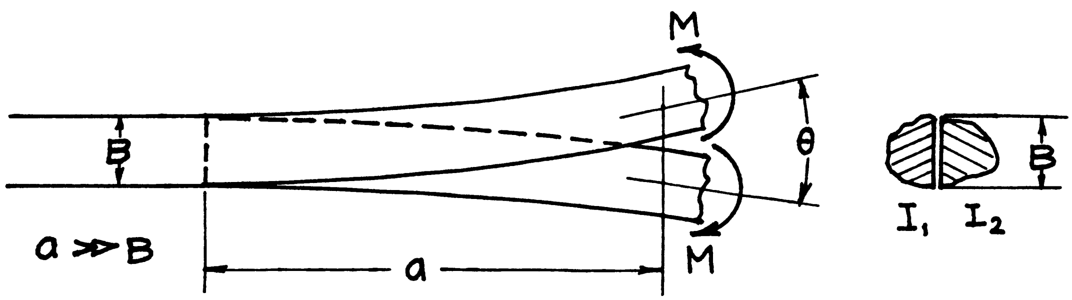
Special cases of **29.5**



$$K_{III} = \frac{1}{\sqrt{1+\nu}} \cdot \frac{Pa}{\sqrt{2BI}}$$

or

$$K_{III} = \frac{1}{\sqrt{1+\nu}} \cdot \frac{3E\Delta}{a^2} \sqrt{\frac{I}{2B}}$$



$$K_{III} = \frac{1}{\sqrt{1+\nu}} \cdot \frac{M}{\sqrt{2BI}}$$

or

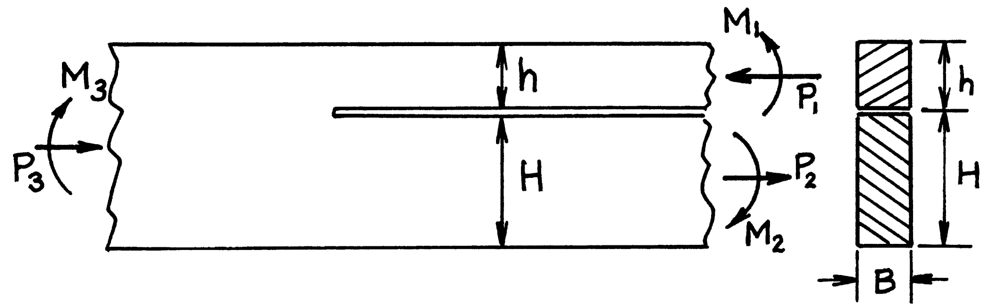
$$K_{III} = \frac{1}{\sqrt{1+\nu}} \cdot \frac{E\theta}{a} \sqrt{\frac{I}{2B}}$$

where $$\frac{1}{I} = \frac{1}{I_1} + \frac{1}{I_2}$$

$I_1, I_2 =$ moment of inertia of each cross section about its neutral axis

Method: Energy Balance
Accuracy: Approximation by Simple Beam Theory
Reference: **Tada 1974**

NOTE: Loads are applied through the shear center of each beam so as to produce bending only with no torsion.



$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II}$$

$$= \frac{1}{E'B^2}\left[\frac{1}{2}\left(\frac{P_1^2}{h} + \frac{P_2^2}{H} + \frac{P_3^2}{h+H}\right) + 6\left(\frac{M_1^2}{h^3} + \frac{M_2^2}{H^3} + \frac{M_3^2}{(h+H)^3}\right)\right]$$

$$K_I = \frac{\left(P/B\right)}{\sqrt{2h}} f(\eta)\cos\omega(\eta) + \frac{\left(M/B\right)}{\sqrt{2h^3}} g(\eta)\sin[\omega(\eta) + \gamma(\eta)]$$

$$K_{II} = \frac{\left(P/B\right)}{\sqrt{2h}} f(\eta)\sin\omega(\eta) - \frac{\left(M/B\right)}{\sqrt{2h^3}} g(\eta)\cos[\omega(\eta) + \gamma(\eta)]$$

where $\eta = h/H, \quad 0 \le \eta \le 1$

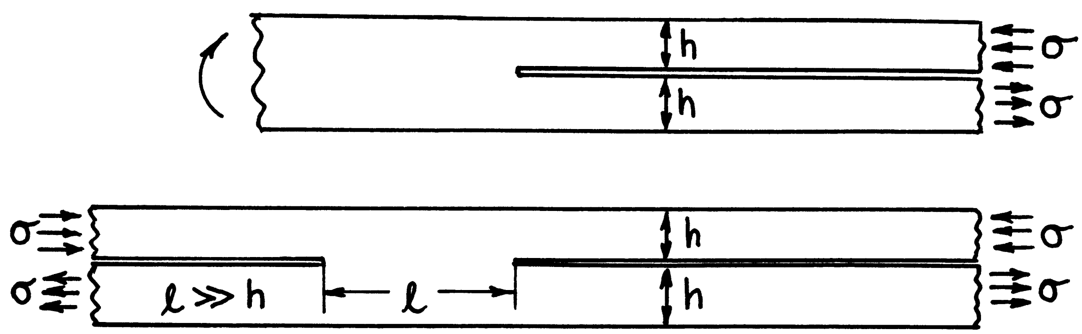
$$P = P_1 - \frac{\eta}{1+\eta}P_3 - \frac{6\eta^2}{(1+\eta)^3}\frac{M_3}{h}, \quad M = M_1 - \left(\frac{\eta}{1+\eta}\right)^3 M_3$$

$$f(\eta) = \left(1 + 4\eta + 6\eta^2 + 3\eta^3\right)^{1/2}, \quad g(\eta) = 2\sqrt{3}\left(1 + \eta^3\right)^{1/2}$$

$$\omega(\eta) = 0.909 - 0.052\eta, \quad \gamma(\eta) = \sin^{-1}\left[\frac{6\eta^2(1+\eta)}{f(\eta)g(\eta)}\right]$$

Method: Energy Balance/Integral Equations
Accuracy: 1%
References: **Suo 1990a; Hutchinson 1992**

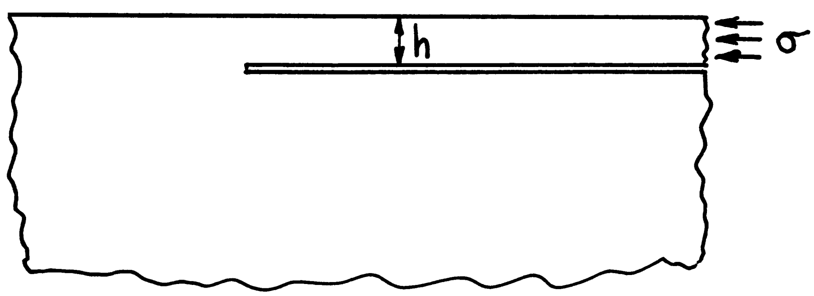
(a)



$$K_I = 0$$

$$K_{II} = \frac{1}{2}\sigma\sqrt{h}$$

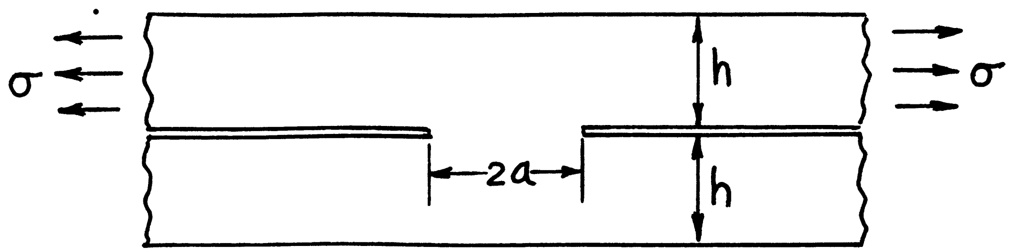
(b)



$$K_I = 0.4347 \ \ \sigma\sqrt{h}$$

$$K_{II} = 0.5578 \ \ \sigma\sqrt{h}$$

Special cases of **29.8**

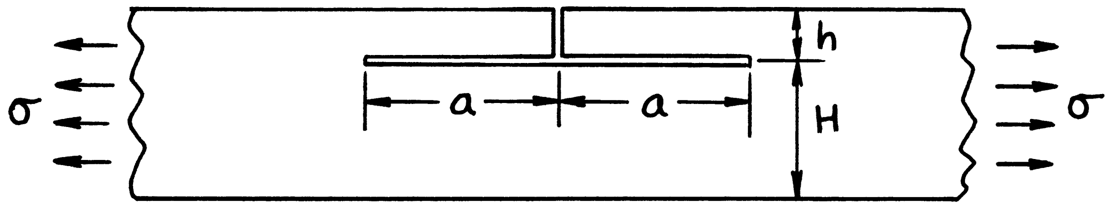


$$K_{II} = \frac{1}{4}\sigma\sqrt{h}\,F(s)$$

$$\text{where} \quad s = \frac{h}{a+h}$$

$$F(s) = \sqrt{1-s}\left(1 + 0.5s + 0.375s^2 - 1.081s^3 + 5.580s^4 - 4.601s^5\right)$$

Methods: Integral Transform Method (Keer; $s \leq 2/3$), Interpolation (Tada; $s > 2/3$)
Accuracy: 1%
References: **Keer 1974, 1989, 1990; Tada 2000**



$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II} = \frac{\sigma^2 h}{2E'} \{f(\eta)\}^2$$

$$\mathcal{G}_I = \cos^2 \omega(\eta) \cdot \mathcal{G}$$

$$\mathcal{G}_{II} = \sin^2 \omega(\eta) \cdot \mathcal{G}$$

$$K_I = \frac{\sigma\sqrt{h}}{\sqrt{2}} f(\eta) \cos \omega(\eta)$$

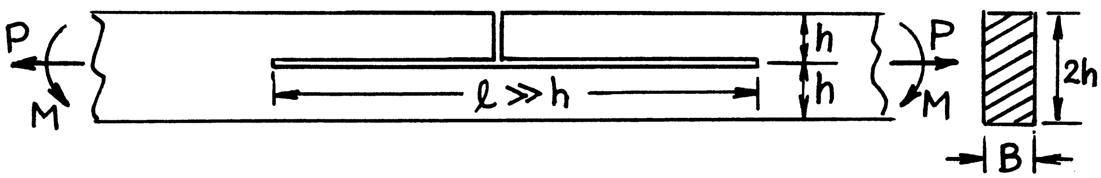
$$K_{II} = \frac{\sigma\sqrt{h}}{\sqrt{2}} f(\eta) \sin \omega(\eta)$$

where $\eta = {}^{h}\!/_{H}$

$$f(\eta) = \left(1 + 4\eta + 6\eta^2 + 3\eta^3\right)^{1/2}$$

$$\omega(\eta) = 0.909 - 0.052\eta$$

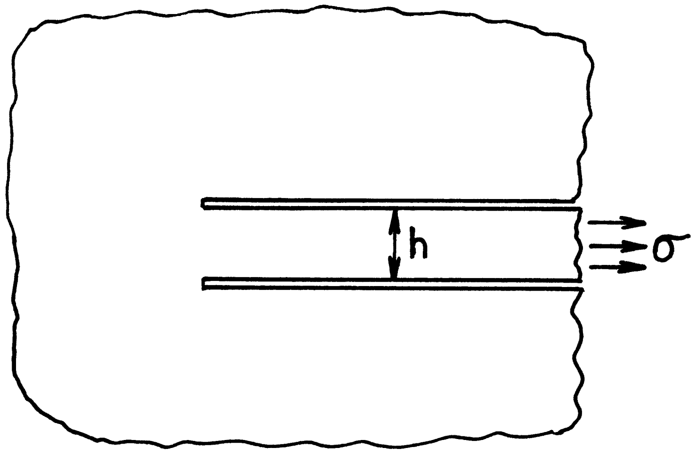
Method: Energy Balance (Special Case of **29.8**)
Accuracy: Better than 1% for ${}^{h}\!/_{a} > 1.5$
Reference: **Suo 1990a**



$$K_I = \frac{\sqrt{3}}{2}\frac{\left(P/B\right)}{\sqrt{h}} + \sqrt{3}\,\frac{\left(M/B\right)}{h^{3/2}}$$

$$K_{II} = \frac{\left(P/B\right)}{\sqrt{h}} + \frac{3}{2}\frac{\left(M/B\right)}{h^{3/2}}$$

Special cases of **29.8** and **29.11**



$$\mathcal{G} = \mathcal{G}_I + \mathcal{G}_{II} = \frac{\sigma^2 h}{4E'}$$

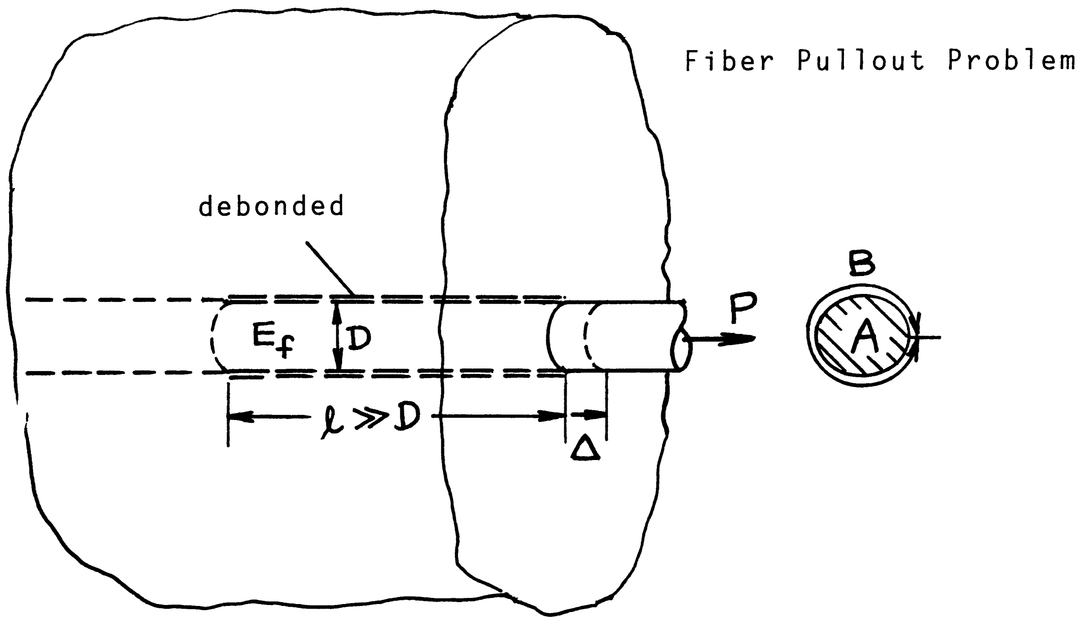
$$\mathcal{G}_I = 0.0904\mathcal{G}$$

$$\mathcal{G}_{II} = 0.9096\mathcal{G}$$

$$K_I = 0.3007 \cdot \frac{1}{2}\sigma\sqrt{h}$$

$$K_{II} = 0.9537 \cdot \frac{1}{2}\sigma\sqrt{h}$$

Method: Energy Balance/Integral Equations
Accuracy: Accurate numerical values
Reference: **Suo 1990b**



$$\mathcal{G}(\simeq \mathcal{G}_{II}) = \frac{P^2}{2E_f A} \cdot \frac{1}{B} = \frac{\sigma^2 A}{2E_f} \cdot \frac{1}{B}$$

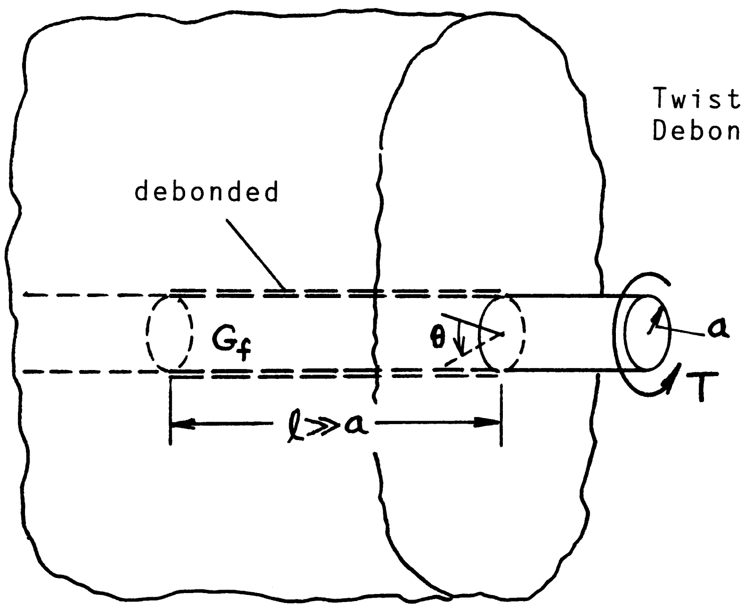
$$= \frac{E_f A \Delta^2}{2\ell^2} \cdot \frac{1}{B}$$

where E_f = Young's modulus of fiber
A = cross sectional area of fiber
B = girth of fiber
ℓ = debonded length
$\sigma = P/A$

For circular fiber of diameter D:

$$\mathcal{G} = \frac{\sigma^2 D}{8E_f} = \frac{E_f D}{8\ell^2}\Delta^2$$

Method: Energy Balance
Accuracy: Approximation by simple tension for fiber
Reference: **Tada 2000**



$$\mathcal{G}(\simeq \mathcal{G}_{III}) = \frac{T^2}{2G_f J}\cdot\frac{1}{B} = \frac{1}{2}G_f J\left(\frac{\theta}{\ell}\right)^2\cdot\frac{1}{B}$$

$$= \frac{\tau^2 a}{8G_f} = \frac{1}{8}G_f a^3\left(\frac{\theta}{\ell}\right)^2$$

where

$G_f =$ shear modulus of fiber

$J =$ polar moment of inertia of fiber $= \dfrac{\pi a^4}{2}$

$B =$ circumference of fiber $= 2\pi a$

$\ell =$ debonded length

$\theta =$ angle of twist

$\tau = \left({}^{T}\!/_{J}\right)a = 2T/(\pi a^3)$

Method: Energy Balance
Accuracy: Approximation by simple torsion of fiber
Reference: **Tada 2000**